

Hedging or Hurting? Geopolitical Risk and the Asymmetric, Regime-Dependent Behaviour of Cryptocurrency Returns

Aswathy Raju^{*1}, Akhil Raju¹, Irala Lokanandha Reddy¹

School of Management Studies, University of Hyderabad, India¹

^{*} Corresponding author at University of Hyderabad, CR Rao Road, Hyderabad Central University Rd, Gachibowli, Hyderabad, Telangana 500046
E-mail 21mbph01@uohyd.ac.in (Aswathy Raju), vakhilraju@gmail.com (Akhil Raju), dr.irala@gmail.com (Irala Lokanandha Reddy)

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E-mail 21mbph01@uohyd.ac.in (Aswathy Raju), vakhilraju@gmail.com (Akhil Raju), dr.irala@gmail.com (Irala Lokanandha Reddy)

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Abstract

This study examines the influence of geopolitical risk (GPR) on cryptocurrency returns using daily global and monthly country-specific data from January 2019 to December 2025. Employing fixed effects panel regression, feasible generalised least squares, heteroskedasticity-based IV estimation, and panel quantile regression across eight major non-stablecoin cryptocurrencies, we find that GPR effects are asymmetric and regime-dependent. GPR is associated with higher returns in bearish market conditions, remains insignificant under normal conditions, and dampens returns in bullish periods. GPR threats exert a consistently stronger negative effect than GPR acts across all market conditions, reflecting the forward-looking nature of investor behaviour in cryptocurrency markets. Country-specific analysis across 44 countries reveals a clear geographic divide: developed European economies including Sweden, Spain, and Finland show significant negative effects consistent with risk-off behaviour, while emerging and high crypto-adoption markets including South Korea, Saudi Arabia, and Vietnam show positive effects consistent with flight-to-crypto behaviour. Argentina shows a notable reversal between historical and recent GPR measures, reflecting its dramatic macroeconomic deterioration during the study period. Quantile-based Granger causality tests confirm that GPR has predictive power over future returns, particularly during bullish market conditions. These findings challenge the unconditional safe-haven characterisation of cryptocurrencies and carry direct implications for portfolio managers, regulators, and investors seeking safe-haven assets in an increasingly uncertain geopolitical environment.

Keywords: *Geopolitical risk, cryptocurrency returns, panel quantile regression, safe-haven, asymmetric effects, country heterogeneity*

JEL Codes: F51; G11; G12; G15; C23

Hedging or Hurting? The Asymmetric Impact of Geopolitical Risk on Crypto Returns

Introduction

Geopolitical crises reshape financial markets in ways that are rarely predictable. When Russia invaded Ukraine in February 2022, Bitcoin fell sharply alongside global equities before recovering within days (Bossman et al., 2023; Boungou & Urom, 2025). When US-China trade tensions escalated in 2019, cryptocurrency markets moved in patterns disconnected from traditional safe havens (Bouri et al., 2022; Long et al., 2022). These events raise a fundamental question that motivates this paper: does geopolitical risk hedge or hurt cryptocurrency returns? The literature on the impact of geopolitical tensions on the cryptocurrency market remains underdeveloped, with limited depth of understanding and no clear consensus in the empirical evidence (Kamal & Wahlstrøm, 2023; Long et al., 2022).

While the impact of GPR on traditional financial assets such as equities and bonds is well documented (Aysan et al., 2019; Bajra et al., 2025; Bekaert et al., 2013; Coën & Desfleurs, 2024; Fang et al., 2024; Iacoviello et al., 2024; Liu et al., 2024), its influence on cryptocurrencies remains far less understood. Some research positions cryptocurrencies as geopolitical safe havens (Bouri et al., 2022; Chibane & Janson, 2025a; Su et al., 2020; Theiri, 2024), whereas other studies report adverse or ambiguous effects (Fereydooni & Hajizadeh, 2024; Hakan Yilmazkuday, 2024; Jana, 2024). Notably, Hakan Yilmazkuday, (2024) finds that the GPR Act has no significant impact on the returns of any of the top ten cryptocurrencies, highlighting the inconsistency in the evidence. In addition, only a limited number of studies examine the cross-asset or temporal variation in the impact of geopolitical tensions, despite the growing interest in the relationship between global instability and financial markets. The nonlinear effects of geopolitical shocks on cryptocurrency returns are also not yet fully understood. Moreover, the country-level heterogeneity in these effects, particularly which

nations' geopolitical tensions exert a stronger influence on cryptocurrency markets, remains an open question in the literature (Almeida et al., 2024; Bouri et al., 2020; Fang et al., 2024; Long et al., 2022). Finally, much of the existing work does not employ sufficiently rigorous or sophisticated methodologies, which limits the ability to establish clear causal relationships.

To address these gaps, this study investigates how global and country-specific geopolitical risk (GPR) influences cryptocurrency returns across the top eight cryptocurrencies and 44 major countries. Using both daily (January 1, 2019, to December 31, 2025) and monthly data (2019m1 to 2025m12), we adopt a comprehensive econometric framework. Baseline effects are estimated using a fixed effects (FE) panel regression model, distributional impacts are explored through quantile regression (QR). To further ensure robust inference, we complement these estimations with feasible generalized least squares (FGLS) to account for heteroskedasticity and serial correlation in panel data (Bai et al., 2021), and Driscoll–Kraay standard errors to address cross-sectional dependence (Hoechle, 2007). Finally, potential endogeneity concerns are mitigated using Lewbel (2012) heteroskedasticity-based instrumental variable approach, which provides valid instruments when external instruments are weak or unavailable (Baum et al., 2012). Together, these methods let us see how GPR affects returns across the whole distribution, while checking that the results hold up under different tests (Balcilar et al., 2018; Jeong et al., 2012; Song & Taamouti, 2021; Tiwari et al., 2019; Umar et al., 2022).

Against this backdrop, we make four contributions to the literature. We provide the first comprehensive distributional evidence on GPR-crypto dynamics using panel quantile regression across nine return quantiles moving beyond the binary safe-haven debate. We disaggregate GPR into acts and threats, showing that anticipated geopolitical tensions exert a stronger and more persistent effect than realised events. We offer country-level analysis across 44 countries, documenting a clear geographic divide between developed European markets and

emerging high crypto-adoption economies. Finally, we address endogeneity through Lewbel (2012) IV estimation, providing causal rather than merely correlational estimates.

This study extends prior work by uncovering patterns missed in traditional models, demonstrating cross-country heterogeneity, and distinguishing between indices, threats, and acts, all of which exert negative impacts, especially in higher quantiles (Balcilar et al., 2018; Liu et al., 2024; Salisu et al., 2022; Umar et al., 2022). To ensure robustness, we complement fixed effects with FGLS, Driscoll–Kraay corrections, and Lewbel IV methods. These results remain consistent across approaches (Kamal & Wahlstrøm, 2023; Long et al., 2022). The findings have direct implications for portfolio managers who allocate to crypto, regulators designing policy responses to geopolitical shocks, and investors seeking safe-haven assets in an increasingly uncertain world. The paper is structured as follows: Section 2 reviews the literature, Section 3 describes the data and methodology, Section 4 presents results, and Section 5 concludes.

2. Literature Review

2.1. Geopolitical Risk and Traditional Financial Markets

The measurement and impact of geopolitical risk on financial markets has gained substantial attention since Caldara & Iacoviello (2022) introduced their text-based GPR index, constructed from the frequency of newspaper articles referencing adverse geopolitical events. Using this index, a large body of research has documented that rising geopolitical tensions systematically suppress equity returns, widen credit spreads, and elevate risk premia across global markets (Aysan et al., 2019; Bekaert et al., 2013; Salisu et al., 2022). Balcilar et al. (2018) find that geopolitical risks significantly affect stock market dynamics in BRICS economies, with asymmetric effects across market regimes. Bossman et al. (2023) document that the Russia-Ukraine conflict generated significant volatility spillovers across EU sectoral stocks, while

Boungou & Urom (2025) show that geopolitical tensions adversely affect bank stock performance through risk premium channels.

In commodity markets, Baur & Smales (2020) demonstrate that gold serves as an effective hedge against geopolitical risk, with the effect concentrated in extreme GPR episodes. Noguera-Santaella (2016) shows that after 2000, geopolitical influences on oil prices became more nuanced. Elsayed & Helmi (2021) document significant volatility transmission across financial markets during geopolitical stress periods. Caldara et al. (2026) further show that geopolitical risks can be inflationary through supply disruption channels. The consistent finding across this literature is that geopolitical shocks act as systematic risk factors that reprice assets and trigger portfolio reallocation typically away from speculative and toward safe-haven instruments.

2.2. Cryptocurrencies as Safe Havens

A growing strand of the literature examines whether cryptocurrencies, particularly Bitcoin, serve as safe-haven assets during periods of financial stress and geopolitical uncertainty. The flight-to-safety hypothesis applied to crypto markets posits that investors may treat decentralised digital assets as alternatives to fiat-based financial systems during geopolitical crises especially in countries with fragile institutions or capital controls (Bouri et al., 2017; Su et al., 2020). Bouri et al. (2017) were among the first to apply wavelet-based quantile-in-quantile regression to show that Bitcoin hedges global uncertainty in bearish market conditions. Su et al. (2020) confirm that Bitcoin can hedge geopolitical event risk using a time-varying framework.

More recent evidence supports this view in specific contexts. Bouri et al. (2022) show that jumps in geopolitical risk have a singular positive effect on Bitcoin returns. Chibane & Janson (2025b) find Bitcoin to be the best safe haven against geopolitical risk among major asset classes. Their (2024) documents significant spillover effects from geopolitical uncertainty to

cryptocurrency markets, with Bitcoin absorbing some of the shock as an alternative asset. These findings draw theoretical support from the speculative asset and monetary sovereignty literature cryptocurrencies operating outside traditional financial infrastructure may attract capital during episodes of geopolitical instability, particularly in countries where traditional banking systems are under stress.

2.3. Contradictory Evidence and Research Gaps

Despite the safe-haven literature, a substantial body of contradictory evidence has emerged. Hakan Yilmazkuday (2024) finds that the GPR Act has no significant impact on the returns of any of the top ten cryptocurrencies, directly challenging the safe-haven narrative. Fereydooni & Hajizadeh (2024) show that safe-haven opportunities for cryptocurrencies in geopolitically risky environments are limited and context-dependent. Jana (2024) finds that metaverse coins are more sensitive to geopolitical risk than traditional crypto assets, suggesting significant heterogeneity within the crypto market itself. Fang et al. (2024) document that GPR affects cryptocurrency market volatility but not necessarily returns in a consistent direction.

The contradictory findings in this literature reflect three underlying limitations. First, most studies rely on linear or mean-based methods OLS, VAR, GARCH, that cannot capture the asymmetric and regime-dependent nature of the GPR-crypto relationship. Crypto markets behave very differently in bearish versus bullish conditions, and averaging over these regimes produces misleading estimates. Second, the literature has focused predominantly on global aggregate GPR, overlooking the heterogeneous effects of country-specific geopolitical tensions (Almeida et al., 2024; Long et al., 2022). Third, causality as distinct from correlation between GPR and crypto returns has received limited attention, particularly across different quantiles of the return distribution (Jeong et al., 2012; Umar et al., 2022).

2.4. Quantile-Based Approaches in Financial Markets

The application of quantile-based methods to study GPR effects has grown rapidly. Eissa et al. (2024) apply quantile regression to study heterogeneous GPR impacts on Middle Eastern stock markets, documenting asymmetric effects across return quantiles. Long et al. (2022) examine whether geopolitical risk is priced in the cross-section of cryptocurrency returns using quantile regression, finding significant effects concentrated in tail quantiles. Umar et al. (2022) apply quantile-on-quantile regression to global asset returns and document that GPR effects vary substantially across both the return and GPR distributions.

In the broader finance literature, Tiwari et al. (2019) use Granger causality in quantiles to study dependence between gold and emerging stock markets. Jeong et al. (2012) develop the nonparametric causality-in-quantiles test that allows testing for causal relationships at each point in the return distribution a critical methodological advance for studying asymmetric GPR transmission. Song & Taamouti (2021) further develop measures of Granger causality in quantiles for practical application. These methodological contributions form the backbone of our empirical framework, which combines panel quantile regression with Lewbel (2012) IV estimation to provide a comprehensive distributional analysis of GPR effects while addressing endogeneity concerns.

3. Data and Methodology

We collected daily cryptocurrency price data from January 1, 2019 to December 31, 2025 from CoinMarketCap.com. We exclude stablecoins (USDT, USDC) as their near-zero returns by design would dilute distributional estimates. Our final sample comprises eight major cryptocurrencies, ADA, BCH, BNB, BTC, DOGE, ETH, TRON, and XRP yielding 20,448 daily observations in a perfectly balanced panel ($8 \text{ tokens} \times 2,556 \text{ days}$). To avoid the confounding effects of the 2018 cryptocurrency market crash, our sample begins in January 2019, capturing the post-crash recovery and subsequent market cycles through December

2025¹. The sample captures key geopolitical episodes including the Russia–Ukraine war (2022), US–China trade tensions, FTX collapse (2022), and Middle East instability (2023–2025). Descriptive statistics confirm the data's suitability for distributional analysis: the average daily return is 0.11% with a standard deviation of 4.89%, and extreme values ranging from −56.1% to +151.6%, confirming the heavy-tailed distribution of cryptocurrency returns and motivating the use of panel quantile regression over mean-based estimators.

The GPR index² is sourced from Caldara & Iacoviello (2022), constructed by counting newspaper articles discussing adverse geopolitical events as a share of total articles, normalised to average 100 during 1985–2019. Three dimensions of the global daily GPR index are used: overall GPR (GPRD), GPR acts (GPRD_ACT) capturing realised events, and GPR threats (GPRD_THREAT) capturing anticipated tensions. These are entered as natural log transformations in all daily regressions to address right skewness. For country-specific analysis, two monthly measures are used: the Recent Country GPR index (GPRC), drawing from 11 recent newspaper sources from 1985 onwards, and the Historical Country GPR index (GPRHC), drawing from historical archives from 1900 onwards. Both are available for 44 countries. GPRC serves as the primary country-level measure given our recent study period, with GPRHC used as a robustness check. Country-specific indices are standardised to mean zero and unit standard deviation prior to estimation to ensure cross-country coefficient comparability. Descriptive statistics for all variables are presented in Table 2. Control variables including log returns of the USD index, CBOE Volatility Index (VIX), S&P 500, NASDAQ, gold, silver, and Vanguard Small-Cap Growth ETF (VBK) are sourced from Yahoo Finance.

¹ (Manahov, 2024)

² <https://www.matteoiacoviello.com/gpr.htm>

3.1. Dependent variable

Following prior literature that examines the impact of geopolitical risk (GPR) on cryptocurrency markets, we use cryptocurrency returns as one of the most widely used outcome variables to measure cryptocurrency performance (Caldara et al., 2026). Cryptocurrency prices are converted into percentage returns by taking the log difference of daily prices and multiplying it by 100. The formula for calculating cryptocurrency returns is given by the following equation:

$$R_{it} = 100 \times \ln\left(\frac{Close_{it}}{Close_{it-1}}\right) \quad (1)$$

3.2. Independent variables

Our primary variable of interest is geopolitical risk (GPR), defined as the “risk associated with wars, terrorist acts, and tensions between states that affect the normal and peaceful course of international relations” (Caldara & Iacoviello, 2022). The GPR index is measured by the frequency of newspaper articles discussing adverse geopolitical events, capturing the intensity of geopolitical concerns reflected in the media. In this study, the key independent variables include the geopolitical threats, geopolitical act, and GPR. The country-specific GPR indices are derived from automated text searches of electronic newspaper archives, identifying articles referencing geopolitical risks. The geopolitical threat index reflects the anticipation or possibility of future geopolitical tensions, while the geopolitical act index measures actual geopolitical events.³ These variables enable us to analyse both the general and country-specific effects of geopolitical risks on cryptocurrency returns.

The regression model we used here is:

$$R_{it} = \alpha + \beta_1, GPR_{it} + \beta_2, Control_{it} + \delta + \varepsilon_{it} \quad (2)$$

³ Refer to this paper for more details (Caldara & Iacoviello, 2022).

Where R_{it} is the outcome variable, GPR_{it} is the GPR variables affecting the return, $Control_{it}$ is a vector that represents various control variables mentioned in section 2.3, δ represents token fixed effects, and ε indicates the error term.

3.3. Panel Quantile Regression

To capture distributional heterogeneity, we employ panel quantile regression following Koenker (2004). We estimate:

$$Q_{\tau}(R_{it} | GPR_{it}, X_{it}) = \alpha_{\tau} + \beta_{\tau} \cdot GPR_{t-1} + \gamma_{\tau} \cdot R_{it-1} + \delta_{\tau} \cdot X_{it} + \lambda_i + \varepsilon_{it} \quad (3)$$

where $Q_{\tau}(\cdot)$ is the τ -th quantile of cryptocurrency returns conditional on the regressors; β_{τ} captures the effect of GPR on returns at quantile τ , which is our main coefficient of interest; and λ_i represents token-specific fixed effects that control for unobserved differences across cryptocurrencies. The model is estimated across nine quantiles ($\tau = 0.10, 0.20, \dots, 0.90$), covering the full distribution of returns from bearish to bullish market conditions. Standard errors are obtained using 100 bootstrap replications to ensure reliability of inference.

3.4. Granger Causality-in-Quantiles Test

Following Umar et al. (2022), we complement the distributional analysis with a quantile-based Granger causality test to examine whether past geopolitical risk predicts cryptocurrency returns at different points of the return distribution. The null hypothesis holds that a change in marginal GPR does not Granger-cause a change in cryptocurrency returns. For each cryptocurrency separately, we estimate quantile regressions at five quantiles ($\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$) including five lags of GPR and three lags of returns as controls. The joint significance of all five GPR lags is tested at each quantile using an F-test. Rejection of the null hypothesis at quantile τ indicates that past GPR has predictive power over returns under that market condition. This approach captures asymmetric and regime-dependent causality that standard linear Granger causality tests cannot detect.

4. Empirical Results

4.1. Summary Statistics

Table 1 presents the variable definitions and Table 2 shows descriptive statistics for all variables used in the analysis. In Table 2, Panel A covers the daily data used in the main panel analysis, Panel B covers the monthly data used in the country-specific analysis, and Panel C presents return statistics by individual cryptocurrency. The average daily return across all tokens is 0.11% with a standard deviation of 4.88%, confirming the well-documented high volatility of cryptocurrency markets. Extreme values ranging from -56.1% to +151.6% confirm the heavy-tailed distribution that motivates the use of panel quantile regression over mean-based estimators. DOGE exhibits the highest kurtosis (123.05) and skewness (5.61), reflecting the extreme price spike in May 2021. The daily GPR index averages 113.69, above the base period average of 100, with GPR threats (mean = 132.66) consistently exceeding GPR acts (mean = 96.67), suggesting that anticipated geopolitical tensions are more persistent than realised events during 2019–2025. Table 3 presents the correlation matrices for daily and monthly data. All three GPR measures show statistically significant negative correlations with cryptocurrency returns, providing preliminary support for the main hypothesis.

[Insert Table 1 and Table 2: Variable Definitions and Descriptive Statistics here]

[Insert Table 3: Correlation Matrices here]

4.2. Baseline Fixed Effects Results

Table 4 presents the baseline fixed effects regression results. The results tell a clear story: geopolitical risk is bad for cryptocurrency returns. Across all three specifications overall GPR, GPR Acts (GPRA), and GPR Threats (GPRT) the effect is negative and statistically significant. When geopolitical tensions rise, crypto investors sell. The magnitude is economically meaningful. A one percent increase in the GPR index is associated with a 0.28 percentage point daily decline in returns ($\beta = -0.00276$, $p < 0.01$) not trivial for an asset class where daily moves

of 5–10% are common. More striking is the difference between threats and acts. GPR Threats dampen returns nearly five times more than GPR Acts (-0.00236 versus -0.000536). Investors do not wait for wars to break out they react the moment tensions escalate. This forward-looking behaviour is consistent with the broader finance literature on uncertainty and asset prices (Caldara & Iacoviello, 2022; Salisu et al., 2022). Among the control variables, the results are intuitive. When the VIX fear index rises, crypto falls confirming that digital assets are not immune to broad market panic. NASDAQ returns are positively and significantly linked to crypto returns ($\beta = 0.649$, $p < 0.01$), reinforcing the well-documented co-movement between cryptocurrencies and technology equities. The USD index exerts a negative effect, consistent with the risk-off narrative. Gold and silver are largely insignificant at the daily frequency, suggesting crypto does not move closely with precious metals a finding relevant to the safe-haven debate. The robustness of these results to alternative estimation methods including Driscoll-Kraay standard errors, FGLS, and Lewbel IV is discussed in Section 4.5.

[Insert Table 4: Baseline Fixed Effects Results here]

Figure 1 presents country-level estimates using the Recent Country GPR index (GPRC) for all 44 countries. The results reveal a clear geographic pattern. Among countries with significant negative effects, Sweden, Spain, and Finland show the strongest negative coefficients at the 1% level, followed by Denmark, Mexico, Hungary, Taiwan, Netherlands, Ukraine, and Switzerland at the 5% level. These are predominantly developed European economies where geopolitical stress triggers risk-off behaviour investors shift away from crypto toward safer traditional assets such as government bonds and gold. In contrast, South Korea and Peru show the strongest positive effects at the 1% level, followed by Saudi Arabia, Vietnam, China, Malaysia, and Philippines at the 5% level. In these markets, rising geopolitical risk drives investors toward cryptocurrencies as an alternative store of value. South Korea is known for very high retail crypto adoption, while Saudi Arabia and Vietnam have fast-growing crypto

markets where traditional safe havens are less accessible. Notably, Argentina shows a significant positive effect when geopolitical risk rises in Argentina, crypto returns increase consistent with flight-to-crypto behaviour driven by chronic currency instability, capital controls, and hyperinflationary pressures during 2019–2025. For a large group of countries including the United States, Russia, France, Canada, and Australia, country-specific GPR has no statistically significant effect, possibly because their geopolitical risk is already captured in the global GPR index.

[Insert Figure 1: Country-Specific GPR Coefficient Plot (GPRC) here]

4.3. Panel Quantile Regression Results

The baseline results mask important heterogeneity across market conditions. Figure 2 presents the panel quantile regression coefficients for overall GPR (Panel A), GPR Acts (GPRA - Panel B), and GPR Threats (GPRT - Panel C) across return quantiles $\tau = 0.10$ to 0.90 . The navy solid line represents the estimated coefficient, and the blue dashed lines represent the 95% confidence interval from 100 bootstrap replications.

Panel A shows that overall GPR exhibits a consistently negative coefficient across all quantiles, confirming the negative relationship between geopolitical risk and cryptocurrency returns throughout the return distribution. At the lowest quantile ($\tau = 0.10$), the coefficient is approximately -0.0015 and relatively small in magnitude. The effect becomes stronger (more negative) at $\tau = 0.20$, then moderates slightly toward the median. From $\tau = 0.80$ onward, the coefficient turns sharply more negative, reaching its strongest value at $\tau = 0.90$ (~ -0.005). This downward-sloping pattern from lower to upper quantiles confirms that GPR's negative effect on returns is most pronounced during bullish market conditions and relatively muted during bearish phases consistent with risk-off behaviour being most active when markets are already performing well.

Panel B shows GPRA follows a distinctly different pattern. At $\tau = 0.10$, the coefficient is positive ($\sim +0.004$), suggesting that realised geopolitical events are associated with higher returns in bearish markets consistent with a flight-to-crypto effect during actual crises. The coefficient then drops sharply, turning negative from $\tau = 0.20$ onward and becoming increasingly negative at upper quantiles, reaching approximately -0.004 at $\tau = 0.90$. This inverted pattern positive in bear markets and negative in bull markets captures the asymmetric role of geopolitical acts: they may attract capital into crypto during downturns but dampen returns during bull phases as investors rotate to safer assets.

Panel C shows GPRT exhibits a consistently negative coefficient across all quantiles, starting at approximately -0.004 at $\tau = 0.10$ and becoming less negative toward the middle quantiles before turning more negative again at upper quantiles. Unlike GPRA which shows a positive coefficient at the lowest quantile, GPRT is negative throughout indicating that the anticipation of geopolitical tensions consistently suppresses returns regardless of market regime. The coefficient is most negative at extreme quantiles ($\tau = 0.10$ and $\tau = 0.90$), with a slight moderation around $\tau = 0.50 - 0.70$. This pattern confirms that geopolitical threats forward-looking investor concerns rather than realised events have a more pervasive and consistent negative effect on cryptocurrency returns, consistent with the broader finding that GPRT exerts a stronger influence than GPRA across our baseline and robustness specifications.

Taken together, the three panels paint a nuanced picture. Cryptocurrencies are neither unconditional safe havens nor simple risk assets their response to geopolitical risk depends critically on what kind of risk it is and what markets are doing at the time. GPR Threats are universally negative across all market conditions, while GPR Acts offer a brief flight-to-crypto effect in the most extreme bear markets before turning negative. This asymmetric, regime-dependent pattern is consistent across all three GPR measures.

[Insert Figure 2: Panel Quantile Regression Coefficient Plots (GPR, GPRA, GPRT) here]

Figure 2: Panel Quantile Regression — Effects of GPR on Cryptocurrency Returns. Panel A = Overall GPR; Panel B = GPR Acts (GPRA); Panel C = GPR Threats (GPRT). Navy solid line = coefficient. Blue dashed lines = 95% confidence interval (100 bootstrap replications). Dashed horizontal line at zero.

4.4. Country-Level Quantile Regression Results

Figure 3 presents country-level quantile regression results across 44 countries using the GPRC index. The results show that the effect of geopolitical risk on crypto returns varies not only across countries but also across market conditions within each country. Among developed European economies, Spain, Finland, Sweden, Denmark, and Netherlands show negative GPR coefficients that are strongest at middle and upper quantiles their geopolitical risk suppresses crypto returns most significantly during normal and bullish market phases. Hungary, Taiwan, Poland, Switzerland, and Ukraine show similar patterns, consistent with risk-off behaviour in these financially developed markets. South Korea and Malaysia show strong positive coefficients particularly at lower and middle quantiles, suggesting that geopolitical risk from these countries boosts crypto demand during bearish and normal market conditions.

Saudi Arabia, Vietnam, and China show positive effects concentrated at lower quantiles, consistent with flight-to-crypto behaviour during bearish market conditions. Argentina shows a particularly interesting pattern strong positive effect at lower quantiles that fade at higher quantiles suggesting that in bear markets, Argentine geopolitical instability drives investors into crypto as an escape from peso depreciation and capital controls, while in bull markets the effect dissipates. For most remaining countries including the United States, Russia, Canada, France, and Australia, coefficients are statistically insignificant across all quantiles, indicating that their country-specific geopolitical risk adds little beyond the global GPR index. As a robustness check, we replicate the analysis using the Historical Country GPR index (GPRHC). The broad geographic pattern remains consistent developed European economies show negative effects and Asian emerging economies show positive effects. One notable exception

is Argentina, which shows a negative effect under GPRHC but a positive effect under GPRC. This reversal reflects Argentina's dramatic macroeconomic deterioration during 2019–2025, including inflation exceeding 200%, severe peso devaluation, and tightening capital controls, which drove a surge in crypto adoption as a store of value.

[Insert Figure 3: Country-wise Quantile Regression Plots (GPRC, 44 countries) here]

4.5. Robustness Checks

The findings survive every stress test we apply. We subject the baseline results to four robustness checks alternative standard error corrections, an alternative estimation method, endogeneity controls, and an alternative GPR measure and the core finding holds in every case.

First, we address cross-sectional dependence and serial correlation using Driscoll-Kraay standard errors (lag = 8). Financial panel data, particularly cryptocurrency returns, are known to exhibit strong cross-sectional dependence when Bitcoin falls, other cryptocurrencies typically follow. Standard fixed effects standard errors would be mis specified in this setting. Driscoll-Kraay standard errors produce heteroskedasticity and autocorrelation consistent estimates that are also robust to cross-sectional dependence (Hoechle, 2007). The baseline findings remain qualitatively unchanged all three GPR measures retain their negative and statistically significant effects confirming that the results are not an artefact of mis specified standard errors.

Second, we estimate the model using Feasible Generalised Least Squares (FGLS), which explicitly models and corrects for panel-specific heteroskedasticity and serial correlation by estimating the error covariance structure before re-estimating the model. Unlike standard OLS or FE which assume homoscedastic errors, FGLS produces more efficient estimates when these assumptions are violated as is clearly the case with cryptocurrency data, where different tokens exhibit very different volatility profiles. The FGLS results confirm the negative and significant

effect of all three GPR measures on cryptocurrency returns, consistent with the baseline Driscoll-Kraay estimates and reinforcing confidence in the main findings.

Third, we address the potential endogeneity of GPR using the Lewbel IV heteroskedasticity-based instrumental variable approach. Endogeneity may arise if cryptocurrency market crashes themselves generate geopolitical anxiety (reverse causality), or if omitted variables simultaneously affect both GPR and returns. The Lewbel IV method exploits heteroskedasticity in the regressors to construct internal instruments without requiring external instruments, which are difficult to identify in this context. The Lewbel IV results confirm the negative and significant effects of all three GPR measures, consistent with the baseline estimates. The magnitudes are broadly similar to the Driscoll-Kraay results, suggesting that endogeneity bias is modest in this setting. This finding reinforces the robustness of the main results and confirms that the negative GPR-crypto relationship is not driven by reverse causality.

Fourth, we replicate the country-specific fixed effects and quantile regression analyses using the Historical Country GPR index (GPRHC) as an alternative measure. GPRHC draws from historical newspaper archives dating back to 1900 using three major newspapers (New York Times, Chicago Tribune, Washington Post), compared to GPRC which draws from 11 recent newspaper sources. The broad geographic pattern remains consistent across both measures developed European economies show predominantly negative effects and Asian emerging markets show positive effects confirming that the country-level findings are not sensitive to the choice of GPR measure. The corresponding coefficient plot and country-level quantile regression plots using GPRHC are presented in Online Appendix Figures A1 and A2 respectively. One notable exception is Argentina, which shows a negative effect under GPRHC but a positive and significant effect under GPRC. This reversal is economically meaningful and reflects Argentina's dramatic macroeconomic deterioration during 2019–2025 including inflation exceeding 200%, severe peso devaluation, and tightening capital controls which drove

a surge in crypto adoption as a store of value during the recent period that was absent in the longer historical pattern.

Taken together, the robustness checks across Driscoll-Kraay standard errors, FGLS, Lewbel IV estimation, and alternative GPR measures confirm that the main findings reflect genuine and stable underlying relationships between geopolitical risk and cryptocurrency returns, rather than methodological artefacts. We also test a lagged specification of GPR the effect disappears entirely, confirming that cryptocurrency markets incorporate geopolitical information instantaneously rather than with delay, consistent with the semi-strong form of market efficiency in digital asset markets. Full robustness results are presented in Online Appendix Table A1.

4.6. Granger Causality-in-Quantiles Results

Table 5 presents the F-statistics and p-values from the quantile-based Granger causality test. Following (Umar et al., 2022), the null hypothesis that a change in marginal GPR does not Granger-cause cryptocurrency returns is tested across five quantiles. Providing further evidence of the asymmetric relationship documented in the panel quantile regression results.

TRON stands out immediately. Across four of five quantiles including both the bearish ($\tau = 0.10$: $F = 4.13$, $p = 0.001$; $\tau = 0.25$: $F = 5.78$, $p = 0.000$) and bullish ($\tau = 0.75$: $F = 6.16$, $p = 0.000$; $\tau = 0.90$: $F = 6.65$, $p = 0.000$) ends of the distribution past GPR significantly predicts TRON returns. No other cryptocurrency shows this breadth of causal response. BTC, ETH, BNB, and ADA show significant causality at upper quantiles only past GPR predicts their returns during bull markets (BTC: $F = 3.86$, $p = 0.002$; ETH: $F = 3.16$, $p = 0.008$; BNB: $F = 3.98$, $p = 0.001$; ADA: $F = 2.31$, $p = 0.042$) but not during downturns. BCH shows a narrower pattern, with causality only at $\tau = 0.75$ ($F = 2.27$, $p = 0.045$). DOGE and XRP show no causality at any quantile. This is consistent with what we know about these assets DOGE moves on

social media sentiment and XRP's price dynamics are dominated by regulatory developments rather than geopolitical fundamentals. The overall picture reinforces the main finding: GPR's relationship with crypto returns has genuine predictive content, particularly when markets are running hot.

[Insert Table 5: Granger Causality-in-Quantiles Results here]

5. Conclusion and Policy Implications

This study provides comprehensive evidence on the asymmetric, nonlinear, and regime-dependent relationship between geopolitical risk and cryptocurrency returns. Using a multi-method panel framework applied to eight major cryptocurrencies across 2019–2025, we find that GPR acts as a hedge in bearish market conditions, is insignificant in normal conditions, and dampens returns in bullish conditions. This pattern is consistent across overall GPR, GPRA, and GPRT, and robust to endogeneity correction via Lewbel (2012) IV estimation.

Disaggregating GPR into threats and acts reveals that threats exert a stronger negative impact than realised events, reflecting investors' forward-looking behaviour. Country-level analysis using the Recent Country GPR index (GPRC) shows the most significant negative effects in Sweden, Spain, and Finland, reflecting risk-off behaviour in developed European financial markets, while South Korea, Saudi Arabia, and Vietnam show significant positive effects consistent with flight-to-crypto behaviour in high crypto-adoption and emerging markets. A robustness check using the Historical Country GPR index (GPRHC) confirms the broad geographic pattern, with Argentina showing a notable reversal from negative to positive under the recent measure a finding consistent with Argentina's dramatic macroeconomic deterioration during the study period. Quantile-based Granger causality tests further confirm that GPR not only affects contemporaneous returns but also has predictive power over future returns, particularly during bullish market conditions, with TRON showing the most pervasive causality and DOGE and XRP showing no causality at any quantile.

These findings carry several practical implications. For portfolio managers, cryptocurrency allocation strategies should be conditioned on prevailing market regimes crypto may serve a useful hedging role during geopolitical crises in bear markets but should be treated as a risk-on asset in bull markets where GPR exposure becomes a drag. For crypto entrepreneurs and token issuers, country-level heterogeneity implies that GPR from major financial hubs poses greater valuation risk than tensions from smaller economies. For policymakers and financial regulators, the findings underscore the need for regime-aware risk assessment frameworks that account for the nonlinear transmission of geopolitical shocks into digital asset markets.

This study has several limitations that open avenues for future research. First, the analysis is restricted to the top eight non-stablecoin cryptocurrencies future research could extend to smaller altcoins, newly emerging tokens, and tokens across different blockchain ecosystems. Second, while Lewbel IV addresses endogeneity, external instruments based on geopolitical event databases could further strengthen causal identification. Third, country-level analysis relies on monthly data, missing intraday dynamics during acute episodes future research could use high-frequency data and event study methods to examine immediate market reactions to specific geopolitical shocks. Fourth, the nonparametric causality-in-quantiles test of (Jeong et al., 2012) could be applied in future work to formally extend the distributional causality evidence provided here.

Table 1: Variable Definitions

Variable	Definition	Frequency	Transformation
ret	Log return of cryptocurrency i on day/month t : $100 \times \ln(Close_{it} / Close_{it-1})$	Daily / Monthly	Log difference
GPRD	Daily Geopolitical Risk Index constructed from automated text searches of 10 major newspapers. Counts articles referencing adverse geopolitical events as a share of total articles, normalised to average 100 during 1985–2019.	Daily	Natural log in regressions
GPRD_ACT	Daily GPR Acts component. Captures newspaper coverage of realised geopolitical events such as terrorist attacks, military conflicts, and wars.	Daily	Natural log in regressions
GPRD_THREAT	Daily GPR Threats component. Captures newspaper coverage of anticipated geopolitical tensions, reflecting forward-looking investor concern about future events.	Daily	Natural log in regressions
GPRC	Recent Country-Specific GPR Index. Constructed by searching for articles mentioning both geopolitical risk keywords and a specific country name in 11 recent newspaper sources. Available for 44 countries. Coverage: 1985–present.	Monthly	Standardised (mean=0, SD=1)
GPRHC	Historical Country-Specific GPR Index. Same methodology as GPRC but uses historical newspaper archives (New York Times, Chicago Tribune, Washington Post). Coverage: 1900–present. Used as robustness check.	Monthly	Standardised (mean=0, SD=1)
ln_volume	Natural log of trading volume in USD. Controls for liquidity and trading activity of each cryptocurrency.	Daily / Monthly	Natural log
ln_marketcap	Natural log of market capitalisation in USD. Controls for the size of each cryptocurrency.	Daily / Monthly	Natural log

ln_cirsupply	Natural log of circulating supply of coins available in the market.	Daily / Monthly	Natural log
usd	US Dollar Index log return. Measures the value of the US dollar against a basket of foreign currencies. A stronger dollar signals risk-off conditions.	Daily / Monthly	Log difference
vix	CBOE Volatility Index log return. Measures expected near-term market volatility, commonly referred to as the fear index.	Daily / Monthly	Log difference
sp500	S&P 500 Index log return. Captures broad US equity market performance.	Daily / Monthly	Log difference
nasdaq	NASDAQ Composite Index log return. Captures technology-heavy equity market performance, closely linked to risk appetite for innovative assets.	Daily / Monthly	Log difference
gold	Gold spot price log return. Benchmark safe-haven asset included to control for flight-to-safety dynamics.	Daily / Monthly	Log difference
silver	Silver spot price log return. Captures precious metals market dynamics as an additional commodity control.	Daily / Monthly	Log difference
vbk	Vanguard Small-Cap Growth ETF log return. Captures small-cap growth equity performance, representing the risk appetite segment most correlated with crypto.	Daily / Monthly	Log difference

Note: The sample covers January 2019 to December 2024. Daily analysis: 8 cryptocurrencies $\times$ 2,556 days = 20,448 observations. Monthly analysis: 8 cryptocurrencies $\times$ 84 months = 672 observations. GPRD, GPRD_ACT, and GPRD_THREAT are entered as natural logs in daily regressions. GPRC and GPRHC are standardised to mean zero and unit standard deviation prior to monthly regressions to ensure cross-country comparability. All financial control variables are log differences of prices or index levels.

Table 2: Descriptive Statistics**Panel A: Daily Data (2019–2025) — N = 20,448 observations**

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
ret	0.0011	0.0488	-0.5613	1.5164	1.7713	67.9315
GPRD	113.6943	60.3724	0.0000	542.2941	1.6330	9.0653
GPRD_ACT	96.6728	67.8753	0.0000	586.2197	1.1963	5.3602
GPRD_THREAT	132.6554	79.3158	0.0000	809.4868	2.2301	13.7965
ln_volume	21.2621	1.8411	16.2145	26.5840	0.0921	2.3848
ln_marketcap	23.8875	1.9963	19.0633	28.5418	0.1305	2.6674
ln_circsupply	21.3114	3.6655	16.6754	25.8480	-0.0709	1.1787
usd	0.0000	0.0039	-0.0214	0.0164	-0.2157	5.9006
vix	-0.0002	0.0716	-0.4424	0.5541	1.2486	11.2631
sp500	0.0005	0.0114	-0.1277	0.0909	-0.7145	22.3280
nasdaq	0.0006	0.0139	-0.1315	0.1148	-0.4527	13.6448
gold	0.0006	0.0094	-0.0591	0.0578	-0.3868	8.0353
silver	0.0008	0.0187	-0.1239	0.1022	-0.3871	9.1732
vbk	0.0004	0.0145	-0.1484	0.1005	-0.7585	14.2798

Note: ret = daily log return ($\times 100$); GPRD = Daily GPR index (base: 1985–2019=100); GPRD_ACT = Daily GPR Acts; GPRD_THREAT = Daily GPR Threats; ln_volume = log trading volume; ln_marketcap = log market capitalisation; ln_circsupply = log circulating supply; usd = USD index log return; vix = VIX log return; sp500 = S&P 500 log return; nasdaq = NASDAQ log return; gold = gold log return; silver = silver log return; vbk = VBK ETF log return. GPR variables are reported in levels; natural log transformations are used in regressions.

Panel B: Monthly Data (2019–2025) — N = 672 observations

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
ret	0.0355	0.2823	-1.1059	2.0724	1.7986	12.2614
GPRC_pooled	0.3261	0.1514	0.1260	1.1183	1.9511	10.4593
GPRHC_pooled	0.2982	0.1205	0.1188	0.7735	0.9806	4.4394
ln_marketCap	23.8991	1.9956	19.2293	28.4653	0.1261	2.6797
ln_circulatingSupply	21.3124	3.6685	16.6784	25.8480	-0.0709	1.1786
ln_volume	23.9967	2.5604	16.4419	28.4929	-0.6280	3.2053
usd	0.0003	0.0195	-0.0513	0.0462	-0.2172	2.7126
vix	-0.0012	0.2455	-0.6143	0.7556	0.1587	3.3542
sp500	0.0112	0.0474	-0.1337	0.1194	-0.6101	3.3979
nasdaq	0.0140	0.0566	-0.1423	0.1436	-0.4697	3.0174
gold	0.0143	0.0391	-0.0717	0.1005	0.3967	2.6834
silver	0.0178	0.0855	-0.1935	0.2659	0.3356	3.2236
vbk	0.0074	0.0625	-0.2060	0.1538	-0.3964	3.8073

Note: GPRC_pooled = cross-country mean of Recent Country GPR (GPRC) across 44 countries; GPRHC_pooled = cross-country mean of Historical Country GPR (GPRHC) across 44 countries. Both are reported in raw form; standardised values (mean=0, SD=1) are used in country-specific regressions to ensure cross-country comparability. Remaining variables as defined in Panel A.

Panel C: Descriptive Statistics by Cryptocurrency — Daily Returns

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
ADA	0.0008	0.0513	-0.5037	0.5384	0.2334	14.0330
BCH	0.0005	0.0508	-0.5613	0.4601	0.0603	19.3867
BNB	0.0019	0.0439	-0.5431	0.5292	-0.1807	27.1677
BTC	0.0012	0.0328	-0.4647	0.1718	-1.0681	21.8768
DOGE	0.0015	0.0662	-0.5151	1.5164	5.6064	123.0470
ETH	0.0012	0.0430	-0.5507	0.2307	-0.9832	17.2469
TRON	0.0010	0.0444	-0.5232	0.6592	0.4428	36.0935
XRP	0.0006	0.0513	-0.5505	0.5486	0.5355	23.5124
Total	0.0011	0.0488	-0.5613	1.5164	1.7713	67.9315

Note: All returns are daily log returns. DOGE exhibits the highest kurtosis (123.05) and skewness (5.61), reflecting the extreme price spike in May 2021. Total row reports pooled statistics across all 8 cryptocurrencies.

Table 3: Correlation Matrix — Daily Data (N = 20,448)

Panel A											
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) ret	1.000										
(2) GPRD	-0.022*	1.000									
(3) GPRD_ACT	-0.012	0.786*	1.000								
(4) GPRD_THREAT	-0.020*	0.919*	0.518*	1.000							
(5) usd	-0.098*	0.031*	0.020*	0.035*	1.000						
(6) vix	-0.248*	-	-	-	0.081*	1.000					
(7) sp500	0.268*	0.015	0.015*	0.009	-	-	1.000				
(8) nasdaq	0.282*	0.000	0.004	-0.006	0.147*	0.731*	-	1.000			
(9) gold	0.068*	0.040*	0.027*	0.035*	-	-0.008	0.082*	0.080*	1.000		
(10) silver	0.105*	0.029*	0.015	0.028*	-	-	0.192*	0.192*	0.756*	1.000	
(11) vbk	0.286*	-0.005	-0.002	-0.011	0.336*	0.115*	-	0.898*	0.899*	0.103*	1.000

Panel B										
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) ret	1.000									
(2) GPRC_pooled	-0.108*	1.000								
(3) GPRHC_pooled	-0.113*	0.875*	1.000							
(4) usd	-0.100*	0.117*	0.122*	1.000						
(5) vix	-0.236*	-0.020	-0.053	0.341*	1.000					
(6) sp500	0.350*	-0.153*	-0.133*	-0.407*	-0.753*	1.000				
(7) nasdaq	0.354*	-0.204*	-0.171*	-0.347*	-0.639*	0.940*	1.000			
(8) gold	0.021	0.112*	0.031	-0.498*	-0.099*	0.210*	0.177*	1.000		
(9) silver	0.059	0.022	-0.016	-0.451*	-0.191*	0.254*	0.239*	0.645*	1.000	
(10) vbk	0.409*	-0.190*	-0.135*	-0.347*	-0.649*	0.891*	0.880*	0.136*	0.182*	1.000

Note: * denotes statistical significance at the 5% level. GPRC_pooled and GPRHC_pooled are cross-country means of Recent and Historical Country GPR indices across 44 countries. Both show significant negative correlation with monthly crypto returns (-0.108^* and -0.113^* respectively), consistent with the main regression findings.

Table 4: Fixed Effects Robust Results

VARIABLES	(1) (1) GPR	(2) (2) GPRA	(3) (3) GPRT
ln_GPR	-0.00276*** (0.000381)		
ln_GPRA		-0.000536** (0.000216)	
ln_GPRT			-0.00236*** (0.000305)
ln_volume	0.00322* (0.00164)	0.00335* (0.00173)	0.00332* (0.00161)
ln_marketcap	-0.000912 (0.00143)	-0.00115 (0.00147)	-0.00105 (0.00141)
ln_cirsupply	0.00356 (0.00649)	0.00224 (0.00699)	0.00258 (0.00686)
usd	-0.523*** (0.0574)	-0.549*** (0.0563)	-0.525*** (0.0569)
vix	-0.0684*** (0.0129)	-0.0680*** (0.0131)	-0.0687*** (0.0129)
sp500	-0.551*** (0.0735)	-0.589*** (0.0768)	-0.552*** (0.0734)
nasdaq	0.649*** (0.0866)	0.664*** (0.0902)	0.650*** (0.0868)
gold	0.0591 (0.0543)	0.0558 (0.0536)	0.0580 (0.0541)
silver	0.0789** (0.0250)	0.0808** (0.0248)	0.0793** (0.0249)
vbk	0.548*** (0.0496)	0.562*** (0.0496)	0.545*** (0.0499)
Constant	-0.110 (0.138)	-0.0894 (0.148)	-0.0894 (0.145)
Observations	17,016	16,632	16,976
R-squared	0.098	0.099	0.098
Number of tokens	8	8	8
Token FE	Yes	Yes	Yes
SE Type	Robust	Robust	Robust

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.10

Robust SE. Token FE. ***, **, * = 1%, 5%, 10%.

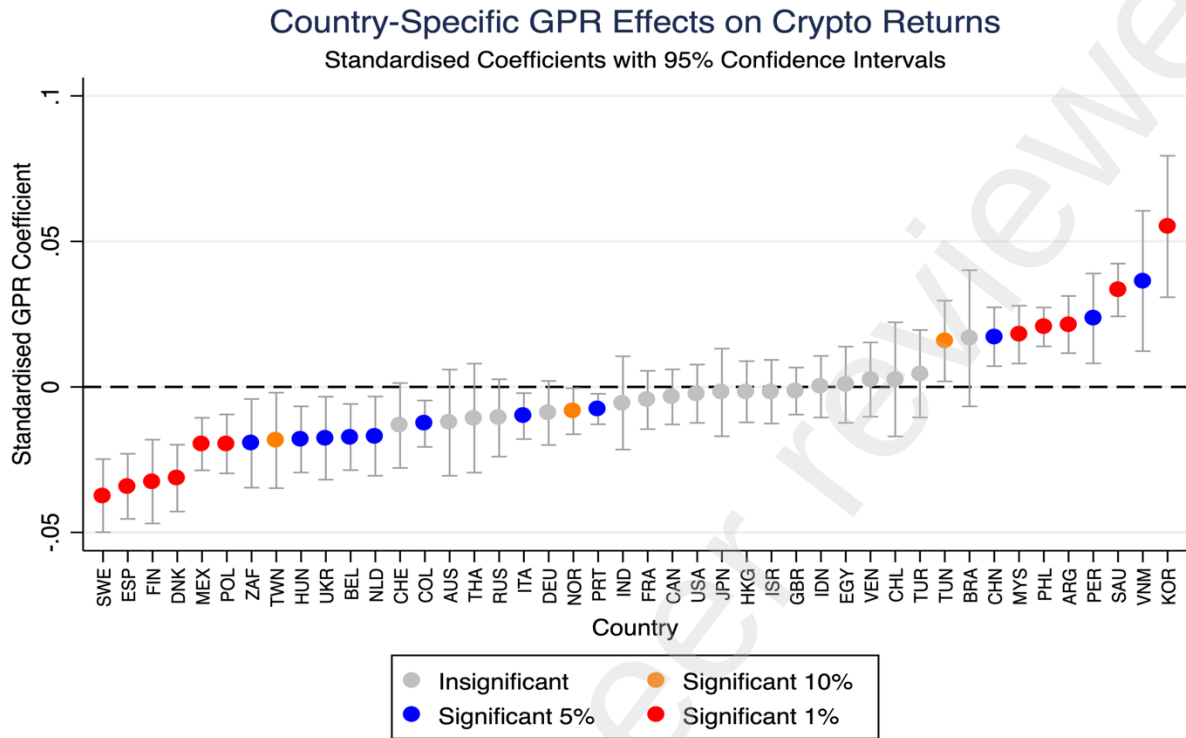
Table 5: Granger Causality-in-Quantiles Test

H₀: A change in marginal GPR does not Granger-cause a change in cryptocurrency returns (Umar et al., 2022)

Cryptocurrency	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
ADA	0.314 [0.905]	0.349 [0.883]	1.885 [0.094*]	2.073 [0.066*]	2.309 [0.042**]
BCH	1.187 [0.313]	0.423 [0.833]	0.399 [0.850]	2.273 [0.045**]	0.914 [0.471]
BNB	3.183 [0.007***]	1.314 [0.255]	0.548 [0.740]	3.984 [0.001***]	3.593 [0.003***]
BTC	0.575 [0.720]	0.390 [0.856]	1.044 [0.390]	3.863 [0.002***]	3.350 [0.005***]
DOGE	0.192 [0.966]	1.235 [0.290]	1.487 [0.191]	1.398 [0.222]	1.204 [0.305]
ETH	0.813 [0.540]	0.291 [0.918]	0.700 [0.623]	3.156 [0.008***]	2.923 [0.012**]
TRON	4.128 [0.001***]	5.781 [0.000***]	0.869 [0.501]	6.158 [0.000***]	6.646 [0.000***]
XRP	1.152 [0.331]	0.873 [0.498]	0.440 [0.821]	1.366 [0.234]	1.151 [0.331]

*Note: Table reports F-statistics with p-values in brackets. F-statistics are from joint significance tests of five GPR lags in quantile regressions estimated separately for each cryptocurrency, following Umar et al. (2022). The null hypothesis is that a change in marginal GPR does not Granger-cause a change in cryptocurrency returns. Three lags of returns are included as controls. $\tau = 0.10$ and 0.25 represent bearish conditions; $\tau = 0.50$ normal conditions; $\tau = 0.75$ and 0.90 bullish conditions. ***, **, * denote significance at 1%, 5%, and 10% levels respectively.*

Figure 1: Impact of Country-level GPR on Crypto Returns



Note: Coefficients sorted by magnitude. Red = 1%, Blue = 5%, Orange = 10%, Grey = Insignificant

Figure 2: Impact of GPR on Crypto Returns (GPR, GPRA, GPRT)

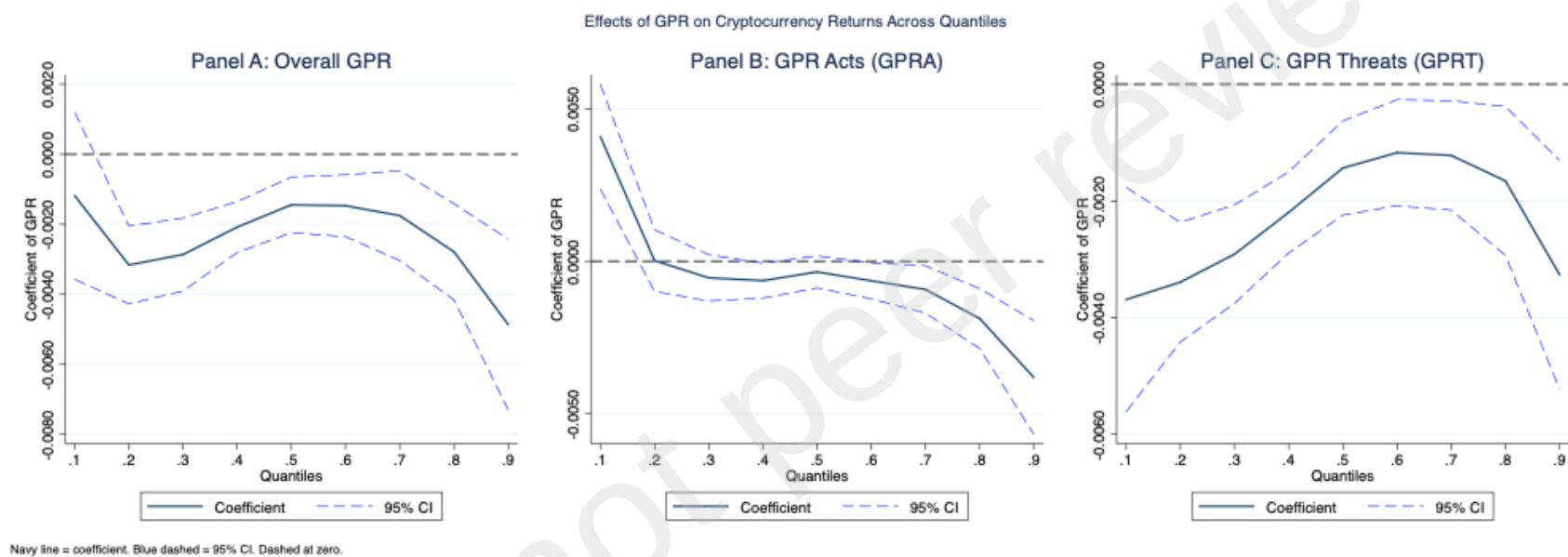
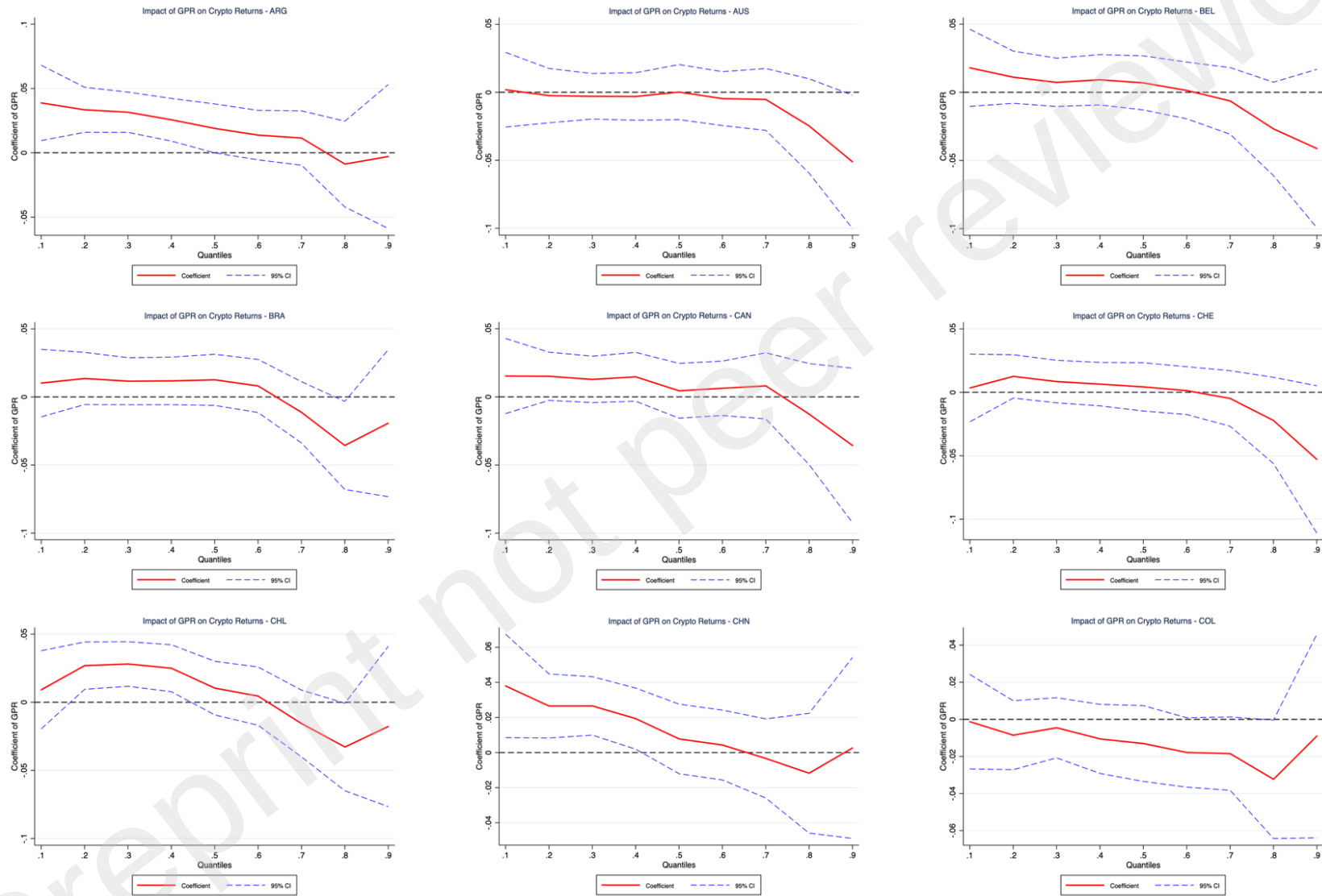
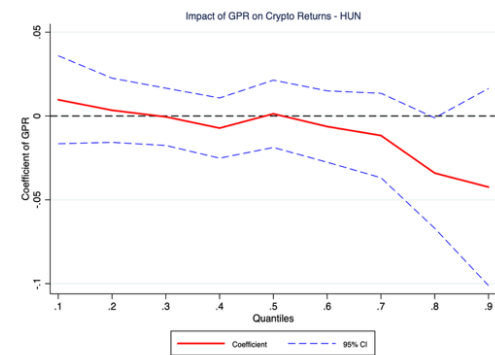
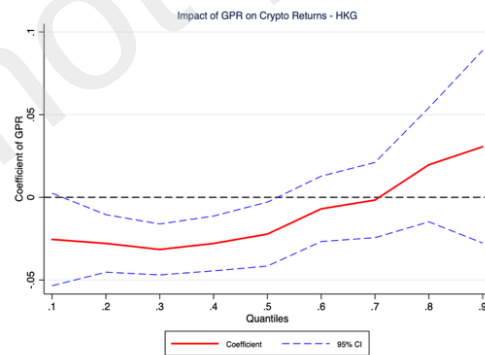
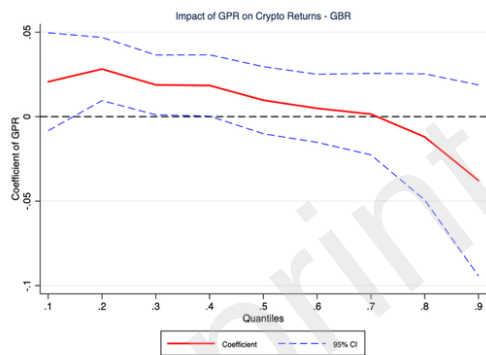
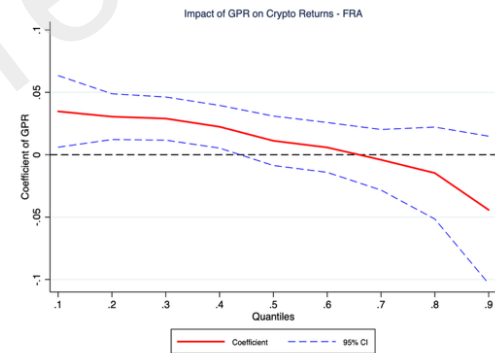
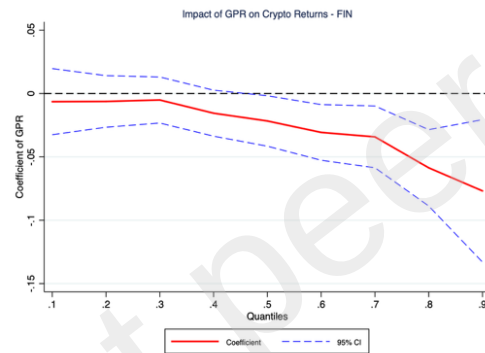
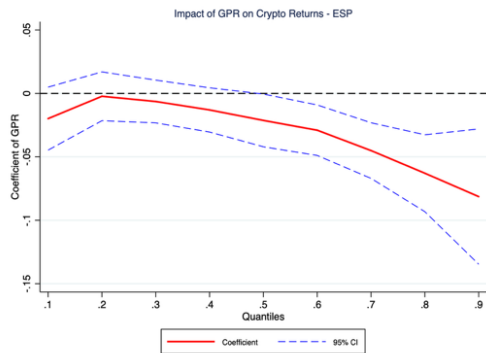
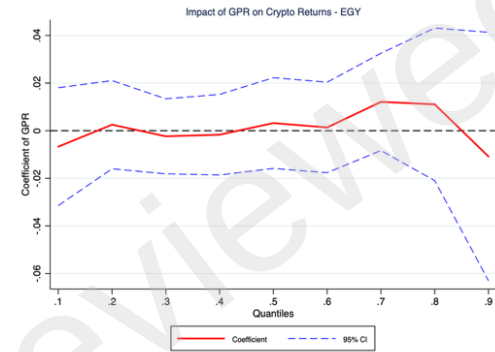
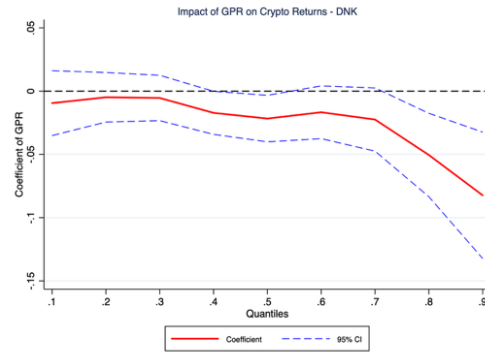
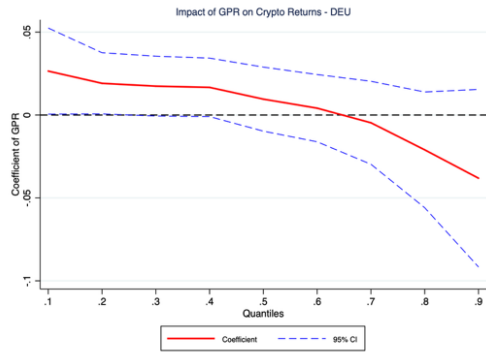
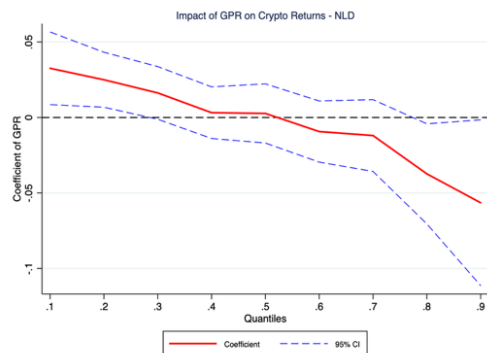
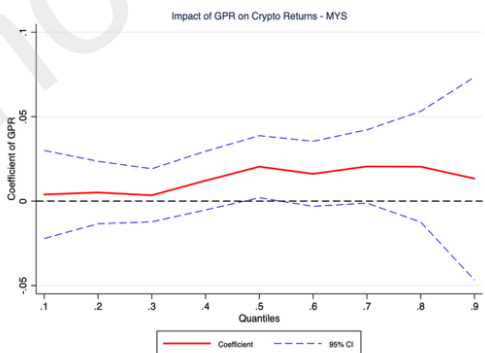
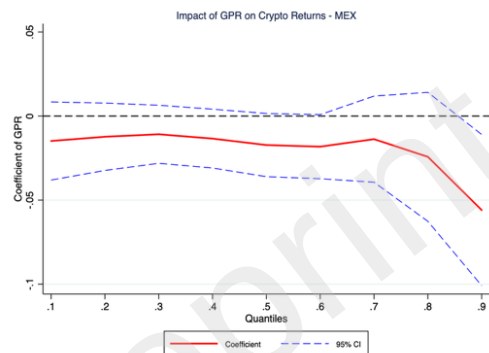
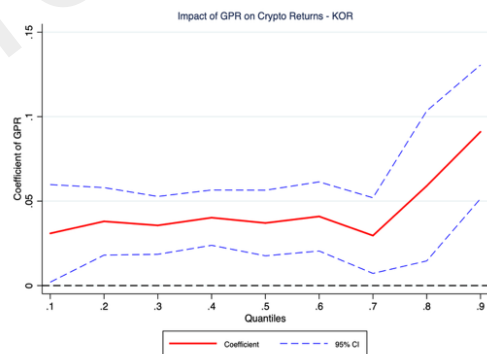
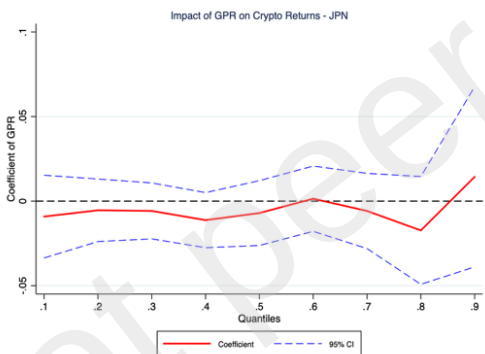
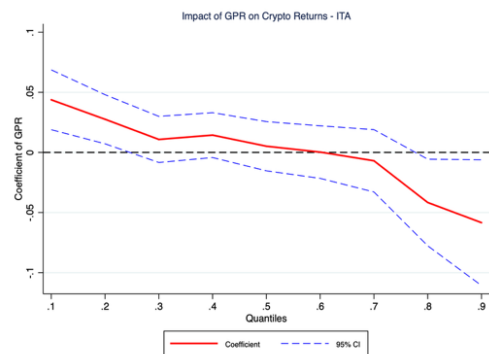
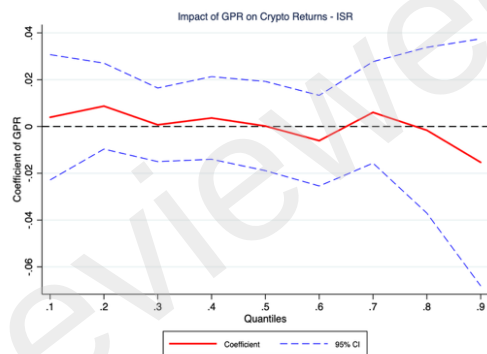
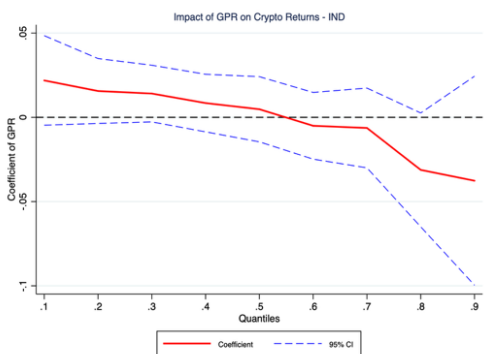
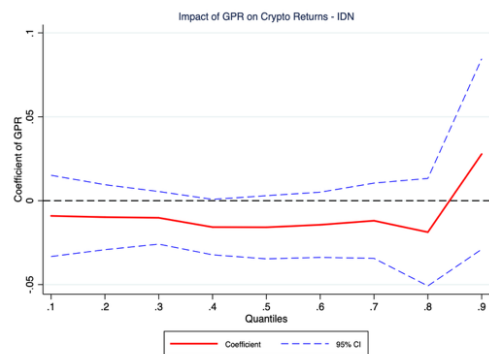
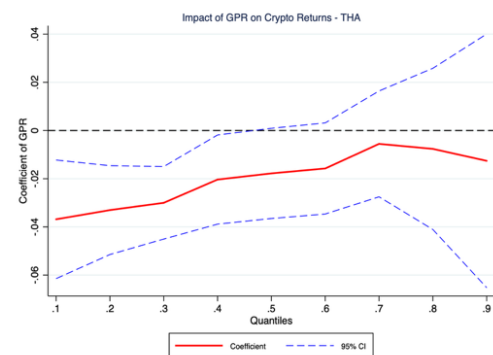
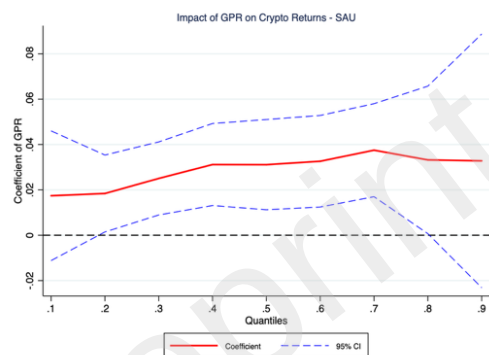
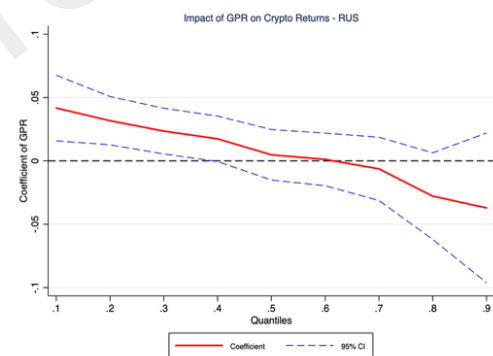
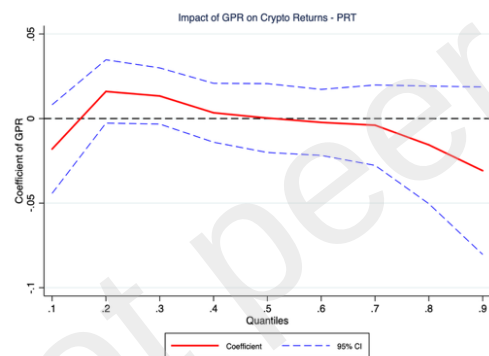
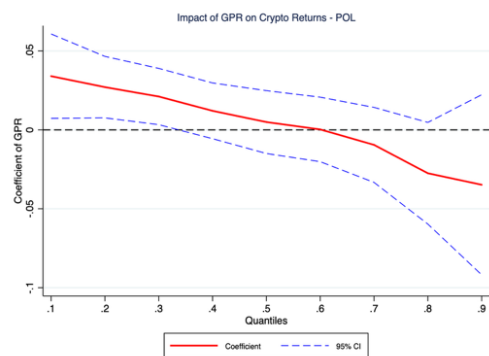
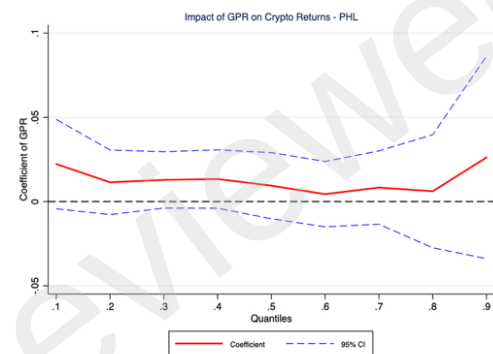
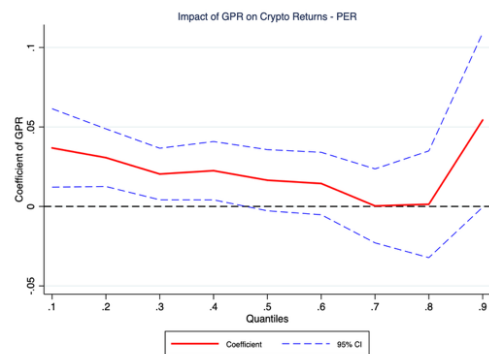


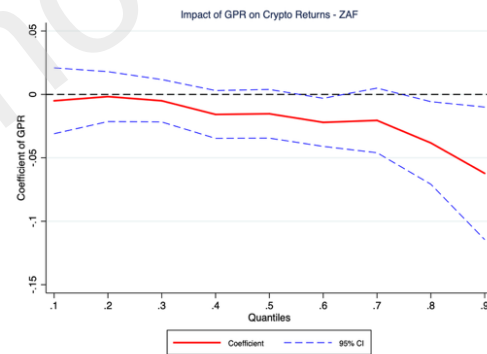
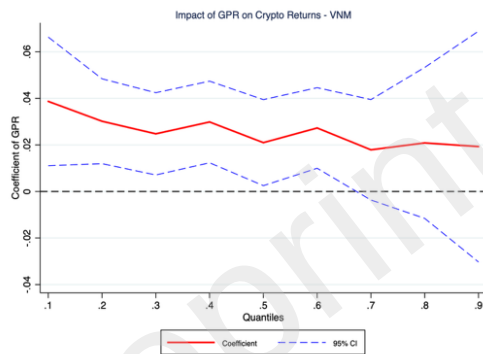
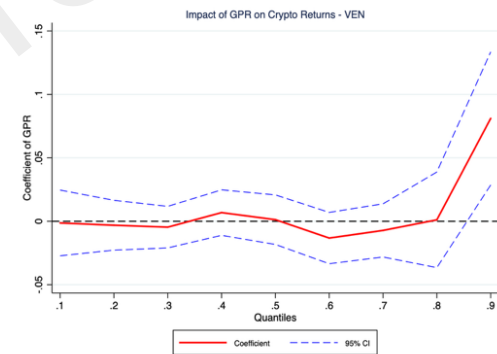
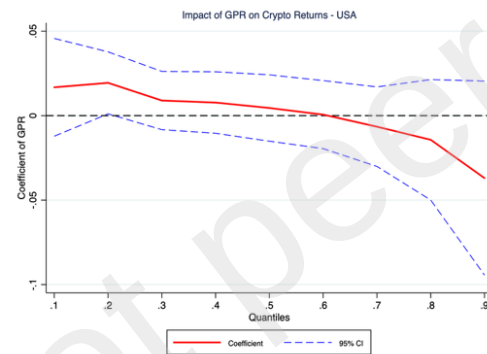
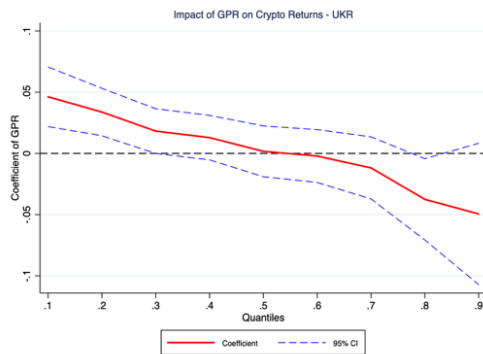
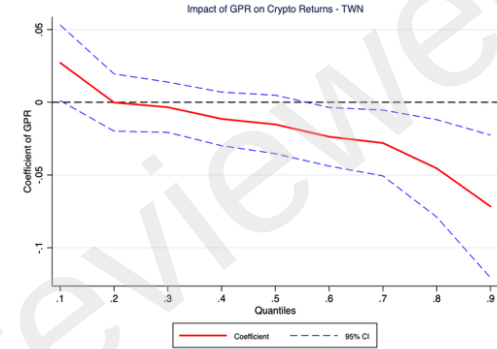
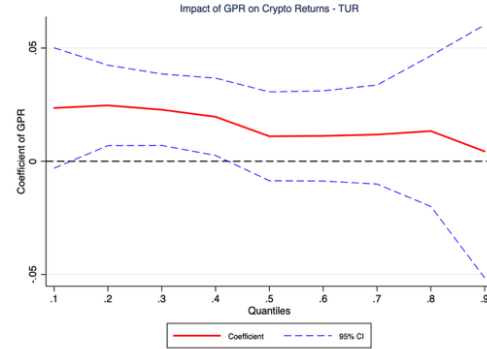
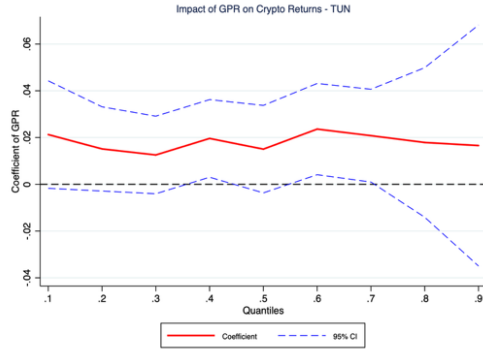
Figure 3: Country-wise QR plots on the impact of GPR on crypto returns (with recent GPR index)











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Hedging or Hurting? Geopolitical Risk and the Asymmetric, Regime-Dependent Behaviour of Cryptocurrency Returns

Abstract

This study examines the influence of geopolitical risk (GPR) on cryptocurrency returns using daily global and monthly country-specific data from January 2019 to December 2025. Employing fixed effects panel regression, feasible generalised least squares, heteroskedasticity-based IV estimation, and panel quantile regression across eight major non-stablecoin cryptocurrencies, we find that GPR effects are asymmetric and regime-dependent. GPR is associated with higher returns in bearish market conditions, remains insignificant under normal conditions, and dampens returns in bullish periods. GPR threats exert a consistently stronger negative effect than GPR acts across all market conditions, reflecting the forward-looking nature of investor behaviour in cryptocurrency markets. Country-specific analysis across 44 countries reveals a clear geographic divide: developed European economies including Sweden, Spain, and Finland show significant negative effects consistent with risk-off behaviour, while emerging and high crypto-adoption markets including South Korea, Saudi Arabia, and Vietnam show positive effects consistent with flight-to-crypto behaviour. Argentina shows a notable reversal between historical and recent GPR measures, reflecting its dramatic macroeconomic deterioration during the study period. Quantile-based Granger causality tests confirm that GPR has predictive power over future returns, particularly during bullish market conditions. These findings challenge the unconditional safe-haven characterisation of cryptocurrencies and carry direct implications for portfolio managers, regulators, and investors seeking safe-haven assets in an increasingly uncertain geopolitical environment.

Keywords: *Geopolitical risk, cryptocurrency returns, panel quantile regression, safe-haven, asymmetric effects, country heterogeneity*

JEL Codes: F51; G11; G12; G15; C23

Hedging or Hurting? The Asymmetric Impact of Geopolitical Risk on Crypto Returns

Introduction

Geopolitical crises reshape financial markets in ways that are rarely predictable. When Russia invaded Ukraine in February 2022, Bitcoin fell sharply alongside global equities before recovering within days (Bossman et al., 2023; Boungou & Urom, 2025). When US-China trade tensions escalated in 2019, cryptocurrency markets moved in patterns disconnected from traditional safe havens (Bouri et al., 2022; Long et al., 2022). These events raise a fundamental question that motivates this paper: does geopolitical risk hedge or hurt cryptocurrency returns? The literature on the impact of geopolitical tensions on the cryptocurrency market remains underdeveloped, with limited depth of understanding and no clear consensus in the empirical evidence (Kamal & Wahlstrøm, 2023; Long et al., 2022).

While the impact of GPR on traditional financial assets such as equities and bonds is well documented (Aysan et al., 2019; Bajra et al., 2025; Bekaert et al., 2013; Coën & Desfleurs, 2024; Fang et al., 2024; Iacoviello et al., 2024; Liu et al., 2024), its influence on cryptocurrencies remains far less understood. Some research positions cryptocurrencies as geopolitical safe havens (Bouri et al., 2022; Chibane & Janson, 2025a; Su et al., 2020; Theiri, 2024), whereas other studies report adverse or ambiguous effects (Fereydooni & Hajizadeh, 2024; Hakan Yilmazkuday, 2024; Jana, 2024). Notably, Hakan Yilmazkuday, (2024) finds that the GPR Act has no significant impact on the returns of any of the top ten cryptocurrencies, highlighting the inconsistency in the evidence. In addition, only a limited number of studies examine the cross-asset or temporal variation in the impact of geopolitical tensions, despite the growing interest in the relationship between global instability and financial markets. The nonlinear effects of geopolitical shocks on cryptocurrency returns are also not yet fully understood. Moreover, the country-level heterogeneity in these effects, particularly which

nations' geopolitical tensions exert a stronger influence on cryptocurrency markets, remains an open question in the literature (Almeida et al., 2024; Bouri et al., 2020; Fang et al., 2024; Long et al., 2022). Finally, much of the existing work does not employ sufficiently rigorous or sophisticated methodologies, which limits the ability to establish clear causal relationships.

To address these gaps, this study investigates how global and country-specific geopolitical risk (GPR) influences cryptocurrency returns across the top eight cryptocurrencies and 44 major countries. Using both daily (January 1, 2019, to December 31, 2025) and monthly data (2019m1 to 2025m12), we adopt a comprehensive econometric framework. Baseline effects are estimated using a fixed effects (FE) panel regression model, distributional impacts are explored through quantile regression (QR). To further ensure robust inference, we complement these estimations with feasible generalized least squares (FGLS) to account for heteroskedasticity and serial correlation in panel data (Bai et al., 2021), and Driscoll–Kraay standard errors to address cross-sectional dependence (Hoechle, 2007). Finally, potential endogeneity concerns are mitigated using Lewbel (2012) heteroskedasticity-based instrumental variable approach, which provides valid instruments when external instruments are weak or unavailable (Baum et al., 2012). Together, these methods let us see how GPR affects returns across the whole distribution, while checking that the results hold up under different tests (Balcilar et al., 2018; Jeong et al., 2012; Song & Taamouti, 2021; Tiwari et al., 2019; Umar et al., 2022).

Against this backdrop, we make four contributions to the literature. We provide the first comprehensive distributional evidence on GPR-crypto dynamics using panel quantile regression across nine return quantiles moving beyond the binary safe-haven debate. We disaggregate GPR into acts and threats, showing that anticipated geopolitical tensions exert a stronger and more persistent effect than realised events. We offer country-level analysis across 44 countries, documenting a clear geographic divide between developed European markets and

emerging high crypto-adoption economies. Finally, we address endogeneity through Lewbel (2012) IV estimation, providing causal rather than merely correlational estimates.

This study extends prior work by uncovering patterns missed in traditional models, demonstrating cross-country heterogeneity, and distinguishing between indices, threats, and acts, all of which exert negative impacts, especially in higher quantiles (Balcilar et al., 2018; Liu et al., 2024; Salisu et al., 2022; Umar et al., 2022). To ensure robustness, we complement fixed effects with FGLS, Driscoll–Kraay corrections, and Lewbel IV methods. These results remain consistent across approaches (Kamal & Wahlstrøm, 2023; Long et al., 2022). The findings have direct implications for portfolio managers who allocate to crypto, regulators designing policy responses to geopolitical shocks, and investors seeking safe-haven assets in an increasingly uncertain world. The paper is structured as follows: Section 2 reviews the literature, Section 3 describes the data and methodology, Section 4 presents results, and Section 5 concludes.

2. Literature Review

2.1. Geopolitical Risk and Traditional Financial Markets

The measurement and impact of geopolitical risk on financial markets has gained substantial attention since Caldara & Iacoviello (2022) introduced their text-based GPR index, constructed from the frequency of newspaper articles referencing adverse geopolitical events. Using this index, a large body of research has documented that rising geopolitical tensions systematically suppress equity returns, widen credit spreads, and elevate risk premia across global markets (Aysan et al., 2019; Bekaert et al., 2013; Salisu et al., 2022). Balcilar et al. (2018) find that geopolitical risks significantly affect stock market dynamics in BRICS economies, with asymmetric effects across market regimes. Bossman et al. (2023) document that the Russia-Ukraine conflict generated significant volatility spillovers across EU sectoral stocks, while

Boungou & Urom (2025) show that geopolitical tensions adversely affect bank stock performance through risk premium channels.

In commodity markets, Baur & Smales (2020) demonstrate that gold serves as an effective hedge against geopolitical risk, with the effect concentrated in extreme GPR episodes. Noguera-Santaella (2016) shows that after 2000, geopolitical influences on oil prices became more nuanced. Elsayed & Helmi (2021) document significant volatility transmission across financial markets during geopolitical stress periods. Caldara et al. (2026) further show that geopolitical risks can be inflationary through supply disruption channels. The consistent finding across this literature is that geopolitical shocks act as systematic risk factors that reprice assets and trigger portfolio reallocation typically away from speculative and toward safe-haven instruments.

2.2. Cryptocurrencies as Safe Havens

A growing strand of the literature examines whether cryptocurrencies, particularly Bitcoin, serve as safe-haven assets during periods of financial stress and geopolitical uncertainty. The flight-to-safety hypothesis applied to crypto markets posits that investors may treat decentralised digital assets as alternatives to fiat-based financial systems during geopolitical crises especially in countries with fragile institutions or capital controls (Bouri et al., 2017; Su et al., 2020). Bouri et al. (2017) were among the first to apply wavelet-based quantile-in-quantile regression to show that Bitcoin hedges global uncertainty in bearish market conditions. Su et al. (2020) confirm that Bitcoin can hedge geopolitical event risk using a time-varying framework.

More recent evidence supports this view in specific contexts. Bouri et al. (2022) show that jumps in geopolitical risk have a singular positive effect on Bitcoin returns. Chibane & Janson (2025b) find Bitcoin to be the best safe haven against geopolitical risk among major asset classes. Their (2024) documents significant spillover effects from geopolitical uncertainty to

cryptocurrency markets, with Bitcoin absorbing some of the shock as an alternative asset. These findings draw theoretical support from the speculative asset and monetary sovereignty literature cryptocurrencies operating outside traditional financial infrastructure may attract capital during episodes of geopolitical instability, particularly in countries where traditional banking systems are under stress.

2.3. Contradictory Evidence and Research Gaps

Despite the safe-haven literature, a substantial body of contradictory evidence has emerged. Hakan Yilmazkuday (2024) finds that the GPR Act has no significant impact on the returns of any of the top ten cryptocurrencies, directly challenging the safe-haven narrative. Fereydooni & Hajizadeh (2024) show that safe-haven opportunities for cryptocurrencies in geopolitically risky environments are limited and context-dependent. Jana (2024) finds that metaverse coins are more sensitive to geopolitical risk than traditional crypto assets, suggesting significant heterogeneity within the crypto market itself. Fang et al. (2024) document that GPR affects cryptocurrency market volatility but not necessarily returns in a consistent direction.

The contradictory findings in this literature reflect three underlying limitations. First, most studies rely on linear or mean-based methods OLS, VAR, GARCH, that cannot capture the asymmetric and regime-dependent nature of the GPR-crypto relationship. Crypto markets behave very differently in bearish versus bullish conditions, and averaging over these regimes produces misleading estimates. Second, the literature has focused predominantly on global aggregate GPR, overlooking the heterogeneous effects of country-specific geopolitical tensions (Almeida et al., 2024; Long et al., 2022). Third, causality as distinct from correlation between GPR and crypto returns has received limited attention, particularly across different quantiles of the return distribution (Jeong et al., 2012; Umar et al., 2022).

2.4. Quantile-Based Approaches in Financial Markets

The application of quantile-based methods to study GPR effects has grown rapidly. Eissa et al. (2024) apply quantile regression to study heterogeneous GPR impacts on Middle Eastern stock markets, documenting asymmetric effects across return quantiles. Long et al. (2022) examine whether geopolitical risk is priced in the cross-section of cryptocurrency returns using quantile regression, finding significant effects concentrated in tail quantiles. Umar et al. (2022) apply quantile-on-quantile regression to global asset returns and document that GPR effects vary substantially across both the return and GPR distributions.

In the broader finance literature, Tiwari et al. (2019) use Granger causality in quantiles to study dependence between gold and emerging stock markets. Jeong et al. (2012) develop the nonparametric causality-in-quantiles test that allows testing for causal relationships at each point in the return distribution a critical methodological advance for studying asymmetric GPR transmission. Song & Taamouti (2021) further develop measures of Granger causality in quantiles for practical application. These methodological contributions form the backbone of our empirical framework, which combines panel quantile regression with Lewbel (2012) IV estimation to provide a comprehensive distributional analysis of GPR effects while addressing endogeneity concerns.

3. Data and Methodology

We collected daily cryptocurrency price data from January 1, 2019 to December 31, 2025 from CoinMarketCap.com. We exclude stablecoins (USDT, USDC) as their near-zero returns by design would dilute distributional estimates. Our final sample comprises eight major cryptocurrencies, ADA, BCH, BNB, BTC, DOGE, ETH, TRON, and XRP yielding 20,448 daily observations in a perfectly balanced panel ($8 \text{ tokens} \times 2,556 \text{ days}$). To avoid the confounding effects of the 2018 cryptocurrency market crash, our sample begins in January 2019, capturing the post-crash recovery and subsequent market cycles through December

2025¹. The sample captures key geopolitical episodes including the Russia–Ukraine war (2022), US–China trade tensions, FTX collapse (2022), and Middle East instability (2023–2025). Descriptive statistics confirm the data's suitability for distributional analysis: the average daily return is 0.11% with a standard deviation of 4.89%, and extreme values ranging from –56.1% to +151.6%, confirming the heavy-tailed distribution of cryptocurrency returns and motivating the use of panel quantile regression over mean-based estimators.

The GPR index² is sourced from Caldara & Iacoviello (2022), constructed by counting newspaper articles discussing adverse geopolitical events as a share of total articles, normalised to average 100 during 1985–2019. Three dimensions of the global daily GPR index are used: overall GPR (GPRD), GPR acts (GPRD_ACT) capturing realised events, and GPR threats (GPRD_THREAT) capturing anticipated tensions. These are entered as natural log transformations in all daily regressions to address right skewness. For country-specific analysis, two monthly measures are used: the Recent Country GPR index (GPRC), drawing from 11 recent newspaper sources from 1985 onwards, and the Historical Country GPR index (GPRHC), drawing from historical archives from 1900 onwards. Both are available for 44 countries. GPRC serves as the primary country-level measure given our recent study period, with GPRHC used as a robustness check. Country-specific indices are standardised to mean zero and unit standard deviation prior to estimation to ensure cross-country coefficient comparability. Descriptive statistics for all variables are presented in Table 2. Control variables including log returns of the USD index, CBOE Volatility Index (VIX), S&P 500, NASDAQ, gold, silver, and Vanguard Small-Cap Growth ETF (VBK) are sourced from Yahoo Finance.

¹ (Manahov, 2024)

² <https://www.matteoiacoviello.com/gpr.htm>

3.1. Dependent variable

Following prior literature that examines the impact of geopolitical risk (GPR) on cryptocurrency markets, we use cryptocurrency returns as one of the most widely used outcome variables to measure cryptocurrency performance (Caldara et al., 2026). Cryptocurrency prices are converted into percentage returns by taking the log difference of daily prices and multiplying it by 100. The formula for calculating cryptocurrency returns is given by the following equation:

$$R_{it} = 100 \times \ln\left(\frac{Close_{it}}{Close_{it-1}}\right) \quad (1)$$

3.2. Independent variables

Our primary variable of interest is geopolitical risk (GPR), defined as the “risk associated with wars, terrorist acts, and tensions between states that affect the normal and peaceful course of international relations” (Caldara & Iacoviello, 2022). The GPR index is measured by the frequency of newspaper articles discussing adverse geopolitical events, capturing the intensity of geopolitical concerns reflected in the media. In this study, the key independent variables include the geopolitical threats, geopolitical act, and GPR. The country-specific GPR indices are derived from automated text searches of electronic newspaper archives, identifying articles referencing geopolitical risks. The geopolitical threat index reflects the anticipation or possibility of future geopolitical tensions, while the geopolitical act index measures actual geopolitical events.³ These variables enable us to analyse both the general and country-specific effects of geopolitical risks on cryptocurrency returns.

The regression model we used here is:

$$R_{it} = \alpha + \beta_1 GPR_{it} + \beta_2 Control_{it} + \delta + \varepsilon_{it} \quad (2)$$

³ Refer to this paper for more details (Caldara & Iacoviello, 2022).

Where R_{it} is the outcome variable, GPR_{it} is the GPR variables affecting the return, $Control_{it}$ is a vector that represents various control variables mentioned in section 2.3, δ represents token fixed effects, and ε indicates the error term.

3.3. Panel Quantile Regression

To capture distributional heterogeneity, we employ panel quantile regression following Koenker (2004). We estimate:

$$Q_{\tau}(R_{it} | GPR_{t-1}, X_{it}) = \alpha_{\tau} + \beta_{\tau} GPR_{t-1} + \gamma_{\tau} R_{it-1} + \delta_{\tau} X_{it} + \lambda_i + \varepsilon_{it} \quad (3)$$

where $Q_{\tau}(\cdot)$ is the τ -th quantile of cryptocurrency returns conditional on the regressors; β_{τ} captures the effect of GPR on returns at quantile τ , which is our main coefficient of interest; and λ_i represents token-specific fixed effects that control for unobserved differences across cryptocurrencies. The model is estimated across nine quantiles ($\tau = 0.10, 0.20, \dots, 0.90$), covering the full distribution of returns from bearish to bullish market conditions. Standard errors are obtained using 100 bootstrap replications to ensure reliability of inference.

3.4. Granger Causality-in-Quantiles Test

Following Umar et al. (2022), we complement the distributional analysis with a quantile-based Granger causality test to examine whether past geopolitical risk predicts cryptocurrency returns at different points of the return distribution. The null hypothesis holds that a change in marginal GPR does not Granger-cause a change in cryptocurrency returns. For each cryptocurrency separately, we estimate quantile regressions at five quantiles ($\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$) including five lags of GPR and three lags of returns as controls. The joint significance of all five GPR lags is tested at each quantile using an F-test. Rejection of the null hypothesis at quantile τ indicates that past GPR has predictive power over returns under that market condition. This approach captures asymmetric and regime-dependent causality that standard linear Granger causality tests cannot detect.

4. Empirical Results

4.1. Summary Statistics

Table 1 presents the variable definitions and Table 2 shows descriptive statistics for all variables used in the analysis. In Table 2, Panel A covers the daily data used in the main panel analysis, Panel B covers the monthly data used in the country-specific analysis, and Panel C presents return statistics by individual cryptocurrency. The average daily return across all tokens is 0.11% with a standard deviation of 4.88%, confirming the well-documented high volatility of cryptocurrency markets. Extreme values ranging from -56.1% to +151.6% confirm the heavy-tailed distribution that motivates the use of panel quantile regression over mean-based estimators. DOGE exhibits the highest kurtosis (123.05) and skewness (5.61), reflecting the extreme price spike in May 2021. The daily GPR index averages 113.69, above the base period average of 100, with GPR threats (mean = 132.66) consistently exceeding GPR acts (mean = 96.67), suggesting that anticipated geopolitical tensions are more persistent than realised events during 2019–2025. Table 3 presents the correlation matrices for daily and monthly data. All three GPR measures show statistically significant negative correlations with cryptocurrency returns, providing preliminary support for the main hypothesis.

[Insert Table 1 and Table 2: Variable Definitions and Descriptive Statistics here]

[Insert Table 3: Correlation Matrices here]

4.2. Baseline Fixed Effects Results

Table 4 presents the baseline fixed effects regression results. The results tell a clear story: geopolitical risk is bad for cryptocurrency returns. Across all three specifications overall GPR, GPR Acts (GPRA), and GPR Threats (GPRT) the effect is negative and statistically significant. When geopolitical tensions rise, crypto investors sell. The magnitude is economically meaningful. A one percent increase in the GPR index is associated with a 0.28 percentage point daily decline in returns ($\beta = -0.00276$, $p < 0.01$) not trivial for an asset class where daily moves

of 5–10% are common. More striking is the difference between threats and acts. GPR Threats dampen returns nearly five times more than GPR Acts (-0.00236 versus -0.000536). Investors do not wait for wars to break out they react the moment tensions escalate. This forward-looking behaviour is consistent with the broader finance literature on uncertainty and asset prices (Caldara & Iacoviello, 2022; Salisu et al., 2022). Among the control variables, the results are intuitive. When the VIX fear index rises, crypto falls confirming that digital assets are not immune to broad market panic. NASDAQ returns are positively and significantly linked to crypto returns ($\beta = 0.649$, $p < 0.01$), reinforcing the well-documented co-movement between cryptocurrencies and technology equities. The USD index exerts a negative effect, consistent with the risk-off narrative. Gold and silver are largely insignificant at the daily frequency, suggesting crypto does not move closely with precious metals a finding relevant to the safe-haven debate. The robustness of these results to alternative estimation methods including Driscoll-Kraay standard errors, FGLS, and Lewbel IV is discussed in Section 4.5.

[Insert Table 4: Baseline Fixed Effects Results here]

Figure 1 presents country-level estimates using the Recent Country GPR index (GPRC) for all 44 countries. The results reveal a clear geographic pattern. Among countries with significant negative effects, Sweden, Spain, and Finland show the strongest negative coefficients at the 1% level, followed by Denmark, Mexico, Hungary, Taiwan, Netherlands, Ukraine, and Switzerland at the 5% level. These are predominantly developed European economies where geopolitical stress triggers risk-off behaviour investors shift away from crypto toward safer traditional assets such as government bonds and gold. In contrast, South Korea and Peru show the strongest positive effects at the 1% level, followed by Saudi Arabia, Vietnam, China, Malaysia, and Philippines at the 5% level. In these markets, rising geopolitical risk drives investors toward cryptocurrencies as an alternative store of value. South Korea is known for very high retail crypto adoption, while Saudi Arabia and Vietnam have fast-growing crypto

markets where traditional safe havens are less accessible. Notably, Argentina shows a significant positive effect when geopolitical risk rises in Argentina, crypto returns increase consistent with flight-to-crypto behaviour driven by chronic currency instability, capital controls, and hyperinflationary pressures during 2019–2025. For a large group of countries including the United States, Russia, France, Canada, and Australia, country-specific GPR has no statistically significant effect, possibly because their geopolitical risk is already captured in the global GPR index.

[Insert Figure 1: Country-Specific GPR Coefficient Plot (GPRC) here]

4.3. Panel Quantile Regression Results

The baseline results mask important heterogeneity across market conditions. Figure 2 presents the panel quantile regression coefficients for overall GPR (Panel A), GPR Acts (GPRA - Panel B), and GPR Threats (GPRT - Panel C) across return quantiles $\tau = 0.10$ to 0.90 . The navy solid line represents the estimated coefficient, and the blue dashed lines represent the 95% confidence interval from 100 bootstrap replications.

Panel A shows that overall GPR exhibits a consistently negative coefficient across all quantiles, confirming the negative relationship between geopolitical risk and cryptocurrency returns throughout the return distribution. At the lowest quantile ($\tau = 0.10$), the coefficient is approximately -0.0015 and relatively small in magnitude. The effect becomes stronger (more negative) at $\tau = 0.20$, then moderates slightly toward the median. From $\tau = 0.80$ onward, the coefficient turns sharply more negative, reaching its strongest value at $\tau = 0.90$ (~ -0.005). This downward-sloping pattern from lower to upper quantiles confirms that GPR's negative effect on returns is most pronounced during bullish market conditions and relatively muted during bearish phases consistent with risk-off behaviour being most active when markets are already performing well.

Panel B shows GPRA follows a distinctly different pattern. At $\tau = 0.10$, the coefficient is positive ($\sim +0.004$), suggesting that realised geopolitical events are associated with higher returns in bearish markets consistent with a flight-to-crypto effect during actual crises. The coefficient then drops sharply, turning negative from $\tau = 0.20$ onward and becoming increasingly negative at upper quantiles, reaching approximately -0.004 at $\tau = 0.90$. This inverted pattern positive in bear markets and negative in bull markets captures the asymmetric role of geopolitical acts: they may attract capital into crypto during downturns but dampen returns during bull phases as investors rotate to safer assets.

Panel C shows GPRT exhibits a consistently negative coefficient across all quantiles, starting at approximately -0.004 at $\tau = 0.10$ and becoming less negative toward the middle quantiles before turning more negative again at upper quantiles. Unlike GPRA which shows a positive coefficient at the lowest quantile, GPRT is negative throughout indicating that the anticipation of geopolitical tensions consistently suppresses returns regardless of market regime. The coefficient is most negative at extreme quantiles ($\tau = 0.10$ and $\tau = 0.90$), with a slight moderation around $\tau = 0.50 - 0.70$. This pattern confirms that geopolitical threats forward-looking investor concerns rather than realised events have a more pervasive and consistent negative effect on cryptocurrency returns, consistent with the broader finding that GPRT exerts a stronger influence than GPRA across our baseline and robustness specifications.

Taken together, the three panels paint a nuanced picture. Cryptocurrencies are neither unconditional safe havens nor simple risk assets their response to geopolitical risk depends critically on what kind of risk it is and what markets are doing at the time. GPR Threats are universally negative across all market conditions, while GPR Acts offer a brief flight-to-crypto effect in the most extreme bear markets before turning negative. This asymmetric, regime-dependent pattern is consistent across all three GPR measures.

[Insert Figure 2: Panel Quantile Regression Coefficient Plots (GPR, GPRA, GPRT) here]

Figure 2: Panel Quantile Regression — Effects of GPR on Cryptocurrency Returns. Panel A = Overall GPR; Panel B = GPR Acts (GPRA); Panel C = GPR Threats (GPRT). Navy solid line = coefficient. Blue dashed lines = 95% confidence interval (100 bootstrap replications). Dashed horizontal line at zero.

4.4. Country-Level Quantile Regression Results

Figure 3 presents country-level quantile regression results across 44 countries using the GPRC index. The results show that the effect of geopolitical risk on crypto returns varies not only across countries but also across market conditions within each country. Among developed European economies, Spain, Finland, Sweden, Denmark, and Netherlands show negative GPR coefficients that are strongest at middle and upper quantiles their geopolitical risk suppresses crypto returns most significantly during normal and bullish market phases. Hungary, Taiwan, Poland, Switzerland, and Ukraine show similar patterns, consistent with risk-off behaviour in these financially developed markets. South Korea and Malaysia show strong positive coefficients particularly at lower and middle quantiles, suggesting that geopolitical risk from these countries boosts crypto demand during bearish and normal market conditions.

Saudi Arabia, Vietnam, and China show positive effects concentrated at lower quantiles, consistent with flight-to-crypto behaviour during bearish market conditions. Argentina shows a particularly interesting pattern strong positive effect at lower quantiles that fade at higher quantiles suggesting that in bear markets, Argentine geopolitical instability drives investors into crypto as an escape from peso depreciation and capital controls, while in bull markets the effect dissipates. For most remaining countries including the United States, Russia, Canada, France, and Australia, coefficients are statistically insignificant across all quantiles, indicating that their country-specific geopolitical risk adds little beyond the global GPR index. As a robustness check, we replicate the analysis using the Historical Country GPR index (GPRHC). The broad geographic pattern remains consistent developed European economies show negative effects and Asian emerging economies show positive effects. One notable exception

is Argentina, which shows a negative effect under GPRHC but a positive effect under GPRC. This reversal reflects Argentina's dramatic macroeconomic deterioration during 2019–2025, including inflation exceeding 200%, severe peso devaluation, and tightening capital controls, which drove a surge in crypto adoption as a store of value.

[Insert Figure 3: Country-wise Quantile Regression Plots (GPRC, 44 countries) here]

4.5. Robustness Checks

The findings survive every stress test we apply. We subject the baseline results to four robustness checks alternative standard error corrections, an alternative estimation method, endogeneity controls, and an alternative GPR measure and the core finding holds in every case.

First, we address cross-sectional dependence and serial correlation using Driscoll-Kraay standard errors (lag = 8). Financial panel data, particularly cryptocurrency returns, are known to exhibit strong cross-sectional dependence when Bitcoin falls, other cryptocurrencies typically follow. Standard fixed effects standard errors would be mis specified in this setting. Driscoll-Kraay standard errors produce heteroskedasticity and autocorrelation consistent estimates that are also robust to cross-sectional dependence (Hoechle, 2007). The baseline findings remain qualitatively unchanged all three GPR measures retain their negative and statistically significant effects confirming that the results are not an artefact of mis specified standard errors.

Second, we estimate the model using Feasible Generalised Least Squares (FGLS), which explicitly models and corrects for panel-specific heteroskedasticity and serial correlation by estimating the error covariance structure before re-estimating the model. Unlike standard OLS or FE which assume homoscedastic errors, FGLS produces more efficient estimates when these assumptions are violated as is clearly the case with cryptocurrency data, where different tokens exhibit very different volatility profiles. The FGLS results confirm the negative and significant

effect of all three GPR measures on cryptocurrency returns, consistent with the baseline Driscoll-Kraay estimates and reinforcing confidence in the main findings.

Third, we address the potential endogeneity of GPR using the Lewbel IV heteroskedasticity-based instrumental variable approach. Endogeneity may arise if cryptocurrency market crashes themselves generate geopolitical anxiety (reverse causality), or if omitted variables simultaneously affect both GPR and returns. The Lewbel IV method exploits heteroskedasticity in the regressors to construct internal instruments without requiring external instruments, which are difficult to identify in this context. The Lewbel IV results confirm the negative and significant effects of all three GPR measures, consistent with the baseline estimates. The magnitudes are broadly similar to the Driscoll-Kraay results, suggesting that endogeneity bias is modest in this setting. This finding reinforces the robustness of the main results and confirms that the negative GPR-crypto relationship is not driven by reverse causality.

Fourth, we replicate the country-specific fixed effects and quantile regression analyses using the Historical Country GPR index (GPRHC) as an alternative measure. GPRHC draws from historical newspaper archives dating back to 1900 using three major newspapers (New York Times, Chicago Tribune, Washington Post), compared to GPRC which draws from 11 recent newspaper sources. The broad geographic pattern remains consistent across both measures developed European economies show predominantly negative effects and Asian emerging markets show positive effects confirming that the country-level findings are not sensitive to the choice of GPR measure. The corresponding coefficient plot and country-level quantile regression plots using GPRHC are presented in Online Appendix Figures A1 and A2 respectively. One notable exception is Argentina, which shows a negative effect under GPRHC but a positive and significant effect under GPRC. This reversal is economically meaningful and reflects Argentina's dramatic macroeconomic deterioration during 2019–2025 including inflation exceeding 200%, severe peso devaluation, and tightening capital controls which drove

a surge in crypto adoption as a store of value during the recent period that was absent in the longer historical pattern.

Taken together, the robustness checks across Driscoll-Kraay standard errors, FGLS, Lewbel IV estimation, and alternative GPR measures confirm that the main findings reflect genuine and stable underlying relationships between geopolitical risk and cryptocurrency returns, rather than methodological artefacts. We also test a lagged specification of GPR the effect disappears entirely, confirming that cryptocurrency markets incorporate geopolitical information instantaneously rather than with delay, consistent with the semi-strong form of market efficiency in digital asset markets. Full robustness results are presented in Online Appendix Table A1.

4.6. Granger Causality-in-Quantiles Results

Table 5 presents the F-statistics and p-values from the quantile-based Granger causality test. Following (Umar et al., 2022), the null hypothesis that a change in marginal GPR does not Granger-cause cryptocurrency returns is tested across five quantiles. Providing further evidence of the asymmetric relationship documented in the panel quantile regression results.

TRON stands out immediately. Across four of five quantiles including both the bearish ($\tau = 0.10$: $F = 4.13$, $p = 0.001$; $\tau = 0.25$: $F = 5.78$, $p = 0.000$) and bullish ($\tau = 0.75$: $F = 6.16$, $p = 0.000$; $\tau = 0.90$: $F = 6.65$, $p = 0.000$) ends of the distribution past GPR significantly predicts TRON returns. No other cryptocurrency shows this breadth of causal response. BTC, ETH, BNB, and ADA show significant causality at upper quantiles only past GPR predicts their returns during bull markets (BTC: $F = 3.86$, $p = 0.002$; ETH: $F = 3.16$, $p = 0.008$; BNB: $F = 3.98$, $p = 0.001$; ADA: $F = 2.31$, $p = 0.042$) but not during downturns. BCH shows a narrower pattern, with causality only at $\tau = 0.75$ ($F = 2.27$, $p = 0.045$). DOGE and XRP show no causality at any quantile. This is consistent with what we know about these assets DOGE moves on

social media sentiment and XRP's price dynamics are dominated by regulatory developments rather than geopolitical fundamentals. The overall picture reinforces the main finding: GPR's relationship with crypto returns has genuine predictive content, particularly when markets are running hot.

[Insert Table 5: Granger Causality-in-Quantiles Results here]

5. Conclusion and Policy Implications

This study provides comprehensive evidence on the asymmetric, nonlinear, and regime-dependent relationship between geopolitical risk and cryptocurrency returns. Using a multi-method panel framework applied to eight major cryptocurrencies across 2019–2025, we find that GPR acts as a hedge in bearish market conditions, is insignificant in normal conditions, and dampens returns in bullish conditions. This pattern is consistent across overall GPR, GPRA, and GPRT, and robust to endogeneity correction via Lewbel (2012) IV estimation.

Disaggregating GPR into threats and acts reveals that threats exert a stronger negative impact than realised events, reflecting investors' forward-looking behaviour. Country-level analysis using the Recent Country GPR index (GPRC) shows the most significant negative effects in Sweden, Spain, and Finland, reflecting risk-off behaviour in developed European financial markets, while South Korea, Saudi Arabia, and Vietnam show significant positive effects consistent with flight-to-crypto behaviour in high crypto-adoption and emerging markets. A robustness check using the Historical Country GPR index (GPRHC) confirms the broad geographic pattern, with Argentina showing a notable reversal from negative to positive under the recent measure a finding consistent with Argentina's dramatic macroeconomic deterioration during the study period. Quantile-based Granger causality tests further confirm that GPR not only affects contemporaneous returns but also has predictive power over future returns, particularly during bullish market conditions, with TRON showing the most pervasive causality and DOGE and XRP showing no causality at any quantile.

These findings carry several practical implications. For portfolio managers, cryptocurrency allocation strategies should be conditioned on prevailing market regimes crypto may serve a useful hedging role during geopolitical crises in bear markets but should be treated as a risk-on asset in bull markets where GPR exposure becomes a drag. For crypto entrepreneurs and token issuers, country-level heterogeneity implies that GPR from major financial hubs poses greater valuation risk than tensions from smaller economies. For policymakers and financial regulators, the findings underscore the need for regime-aware risk assessment frameworks that account for the nonlinear transmission of geopolitical shocks into digital asset markets.

This study has several limitations that open avenues for future research. First, the analysis is restricted to the top eight non-stablecoin cryptocurrencies future research could extend to smaller altcoins, newly emerging tokens, and tokens across different blockchain ecosystems. Second, while Lewbel IV addresses endogeneity, external instruments based on geopolitical event databases could further strengthen causal identification. Third, country-level analysis relies on monthly data, missing intraday dynamics during acute episodes future research could use high-frequency data and event study methods to examine immediate market reactions to specific geopolitical shocks. Fourth, the nonparametric causality-in-quantiles test of (Jeong et al., 2012) could be applied in future work to formally extend the distributional causality evidence provided here.

Table 1: Variable Definitions

Variable	Definition	Frequency	Transformation
ret	Log return of cryptocurrency i on day/month t : $100 \times \ln(Close_{it} / Close_{it-1})$	Daily / Monthly	Log difference
GPRD	Daily Geopolitical Risk Index constructed from automated text searches of 10 major newspapers. Counts articles referencing adverse geopolitical events as a share of total articles, normalised to average 100 during 1985–2019.	Daily	Natural log in regressions
GPRD_ACT	Daily GPR Acts component. Captures newspaper coverage of realised geopolitical events such as terrorist attacks, military conflicts, and wars.	Daily	Natural log in regressions
GPRD_THREAT	Daily GPR Threats component. Captures newspaper coverage of anticipated geopolitical tensions, reflecting forward-looking investor concern about future events.	Daily	Natural log in regressions
GPRC	Recent Country-Specific GPR Index. Constructed by searching for articles mentioning both geopolitical risk keywords and a specific country name in 11 recent newspaper sources. Available for 44 countries. Coverage: 1985–present.	Monthly	Standardised (mean=0, SD=1)
GPRHC	Historical Country-Specific GPR Index. Same methodology as GPRC but uses historical newspaper archives (New York Times, Chicago Tribune, Washington Post). Coverage: 1900–present. Used as robustness check.	Monthly	Standardised (mean=0, SD=1)
ln_volume	Natural log of trading volume in USD. Controls for liquidity and trading activity of each cryptocurrency.	Daily / Monthly	Natural log
ln_marketcap	Natural log of market capitalisation in USD. Controls for the size of each cryptocurrency.	Daily / Monthly	Natural log

ln_cirsupply	Natural log of circulating supply of coins available in the market.	Daily / Monthly	Natural log
usd	US Dollar Index log return. Measures the value of the US dollar against a basket of foreign currencies. A stronger dollar signals risk-off conditions.	Daily / Monthly	Log difference
vix	CBOE Volatility Index log return. Measures expected near-term market volatility, commonly referred to as the fear index.	Daily / Monthly	Log difference
sp500	S&P 500 Index log return. Captures broad US equity market performance.	Daily / Monthly	Log difference
nasdaq	NASDAQ Composite Index log return. Captures technology-heavy equity market performance, closely linked to risk appetite for innovative assets.	Daily / Monthly	Log difference
gold	Gold spot price log return. Benchmark safe-haven asset included to control for flight-to-safety dynamics.	Daily / Monthly	Log difference
silver	Silver spot price log return. Captures precious metals market dynamics as an additional commodity control.	Daily / Monthly	Log difference
vbk	Vanguard Small-Cap Growth ETF log return. Captures small-cap growth equity performance, representing the risk appetite segment most correlated with crypto.	Daily / Monthly	Log difference

Note: The sample covers January 2019 to December 2024. Daily analysis: 8 cryptocurrencies $\times$ 2,556 days = 20,448 observations. Monthly analysis: 8 cryptocurrencies $\times$ 84 months = 672 observations. GPRD, GPRD_ACT, and GPRD_THREAT are entered as natural logs in daily regressions. GPRC and GPRHC are standardised to mean zero and unit standard deviation prior to monthly regressions to ensure cross-country comparability. All financial control variables are log differences of prices or index levels.

Table 2: Descriptive Statistics**Panel A: Daily Data (2019–2025) — N = 20,448 observations**

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
ret	0.0011	0.0488	-0.5613	1.5164	1.7713	67.9315
GPRD	113.6943	60.3724	0.0000	542.2941	1.6330	9.0653
GPRD_ACT	96.6728	67.8753	0.0000	586.2197	1.1963	5.3602
GPRD_THREAT	132.6554	79.3158	0.0000	809.4868	2.2301	13.7965
ln_volume	21.2621	1.8411	16.2145	26.5840	0.0921	2.3848
ln_marketcap	23.8875	1.9963	19.0633	28.5418	0.1305	2.6674
ln_circsupply	21.3114	3.6655	16.6754	25.8480	-0.0709	1.1787
usd	0.0000	0.0039	-0.0214	0.0164	-0.2157	5.9006
vix	-0.0002	0.0716	-0.4424	0.5541	1.2486	11.2631
sp500	0.0005	0.0114	-0.1277	0.0909	-0.7145	22.3280
nasdaq	0.0006	0.0139	-0.1315	0.1148	-0.4527	13.6448
gold	0.0006	0.0094	-0.0591	0.0578	-0.3868	8.0353
silver	0.0008	0.0187	-0.1239	0.1022	-0.3871	9.1732
vbk	0.0004	0.0145	-0.1484	0.1005	-0.7585	14.2798

Note: ret = daily log return ($\times 100$); GPRD = Daily GPR index (base: 1985–2019=100); GPRD_ACT = Daily GPR Acts; GPRD_THREAT = Daily GPR Threats; ln_volume = log trading volume; ln_marketcap = log market capitalisation; ln_circsupply = log circulating supply; usd = USD index log return; vix = VIX log return; sp500 = S&P 500 log return; nasdaq = NASDAQ log return; gold = gold log return; silver = silver log return; vbk = VBK ETF log return. GPR variables are reported in levels; natural log transformations are used in regressions.

Panel B: Monthly Data (2019–2025) — N = 672 observations

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
ret	0.0355	0.2823	-1.1059	2.0724	1.7986	12.2614
GPRC_pooled	0.3261	0.1514	0.1260	1.1183	1.9511	10.4593
GPRHC_pooled	0.2982	0.1205	0.1188	0.7735	0.9806	4.4394
ln_marketCap	23.8991	1.9956	19.2293	28.4653	0.1261	2.6797
ln_circulatingSupply	21.3124	3.6685	16.6784	25.8480	-0.0709	1.1786
ln_volume	23.9967	2.5604	16.4419	28.4929	-0.6280	3.2053
usd	0.0003	0.0195	-0.0513	0.0462	-0.2172	2.7126
vix	-0.0012	0.2455	-0.6143	0.7556	0.1587	3.3542
sp500	0.0112	0.0474	-0.1337	0.1194	-0.6101	3.3979
nasdaq	0.0140	0.0566	-0.1423	0.1436	-0.4697	3.0174
gold	0.0143	0.0391	-0.0717	0.1005	0.3967	2.6834
silver	0.0178	0.0855	-0.1935	0.2659	0.3356	3.2236
vbk	0.0074	0.0625	-0.2060	0.1538	-0.3964	3.8073

Note: GPRC_pooled = cross-country mean of Recent Country GPR (GPRC) across 44 countries; GPRHC_pooled = cross-country mean of Historical Country GPR (GPRHC) across 44 countries. Both are reported in raw form; standardised values (mean=0, SD=1) are used in country-specific regressions to ensure cross-country comparability. Remaining variables as defined in Panel A.

Panel C: Descriptive Statistics by Cryptocurrency — Daily Returns

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
ADA	0.0008	0.0513	-0.5037	0.5384	0.2334	14.0330
BCH	0.0005	0.0508	-0.5613	0.4601	0.0603	19.3867
BNB	0.0019	0.0439	-0.5431	0.5292	-0.1807	27.1677
BTC	0.0012	0.0328	-0.4647	0.1718	-1.0681	21.8768
DOGE	0.0015	0.0662	-0.5151	1.5164	5.6064	123.0470
ETH	0.0012	0.0430	-0.5507	0.2307	-0.9832	17.2469
TRON	0.0010	0.0444	-0.5232	0.6592	0.4428	36.0935
XRP	0.0006	0.0513	-0.5505	0.5486	0.5355	23.5124
Total	0.0011	0.0488	-0.5613	1.5164	1.7713	67.9315

Note: All returns are daily log returns. DOGE exhibits the highest kurtosis (123.05) and skewness (5.61), reflecting the extreme price spike in May 2021. Total row reports pooled statistics across all 8 cryptocurrencies.

Table 3: Correlation Matrix — Daily Data (N = 20,448)

Panel A											
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) ret	1.000										
(2) GPRD	-0.022*	1.000									
(3) GPRD_ACT	-0.012	0.786*	1.000								
(4) GPRD_THREAT	-0.020*	0.919*	0.518*	1.000							
(5) usd	-0.098*	0.031*	0.020*	0.035*	1.000						
(6) vix	-0.248*	-	-	-	0.081*	1.000					
(7) sp500	0.268*	0.015	0.015*	0.009	-	-	1.000				
(8) nasdaq	0.282*	0.000	0.004	-0.006	-	-	0.950*	1.000			
(9) gold	0.068*	0.040*	0.027*	0.035*	-	-0.008	0.082*	0.080*	1.000		
(10) silver	0.105*	0.029*	0.015	0.028*	-	-	0.192*	0.192*	0.756*	1.000	
(11) vbk	0.286*	-0.005	-0.002	-0.011	-	-	0.898*	0.899*	0.103*	0.215*	1.000

Panel B										
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) ret	1.000									
(2) GPRC_pooled	-0.108*	1.000								
(3) GPRHC_pooled	-0.113*	0.875*	1.000							
(4) usd	-0.100*	0.117*	0.122*	1.000						
(5) vix	-0.236*	-0.020	-0.053	0.341*	1.000					
(6) sp500	0.350*	-0.153*	-0.133*	-0.407*	-0.753*	1.000				
(7) nasdaq	0.354*	-0.204*	-0.171*	-0.347*	-0.639*	0.940*	1.000			
(8) gold	0.021	0.112*	0.031	-0.498*	-0.099*	0.210*	0.177*	1.000		
(9) silver	0.059	0.022	-0.016	-0.451*	-0.191*	0.254*	0.239*	0.645*	1.000	
(10) vbk	0.409*	-0.190*	-0.135*	-0.347*	-0.649*	0.891*	0.880*	0.136*	0.182*	1.000

Note: * denotes statistical significance at the 5% level. GPRC_pooled and GPRHC_pooled are cross-country means of Recent and Historical Country GPR indices across 44 countries. Both show significant negative correlation with monthly crypto returns (-0.108^* and -0.113^* respectively), consistent with the main regression findings.

Table 4: Fixed Effects Robust Results

VARIABLES	(1) (1) GPR	(2) (2) GPRA	(3) (3) GPRT
ln_GPR	-0.00276*** (0.000381)		
ln_GPRA		-0.000536** (0.000216)	
ln_GPRT			-0.00236*** (0.000305)
ln_volume	0.00322* (0.00164)	0.00335* (0.00173)	0.00332* (0.00161)
ln_marketcap	-0.000912 (0.00143)	-0.00115 (0.00147)	-0.00105 (0.00141)
ln_cirsupply	0.00356 (0.00649)	0.00224 (0.00699)	0.00258 (0.00686)
usd	-0.523*** (0.0574)	-0.549*** (0.0563)	-0.525*** (0.0569)
vix	-0.0684*** (0.0129)	-0.0680*** (0.0131)	-0.0687*** (0.0129)
sp500	-0.551*** (0.0735)	-0.589*** (0.0768)	-0.552*** (0.0734)
nasdaq	0.649*** (0.0866)	0.664*** (0.0902)	0.650*** (0.0868)
gold	0.0591 (0.0543)	0.0558 (0.0536)	0.0580 (0.0541)
silver	0.0789** (0.0250)	0.0808** (0.0248)	0.0793** (0.0249)
vbk	0.548*** (0.0496)	0.562*** (0.0496)	0.545*** (0.0499)
Constant	-0.110 (0.138)	-0.0894 (0.148)	-0.0894 (0.145)
Observations	17,016	16,632	16,976
R-squared	0.098	0.099	0.098
Number of tokens	8	8	8
Token FE	Yes	Yes	Yes
SE Type	Robust	Robust	Robust

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.10

Robust SE. Token FE. ***, **, * = 1%, 5%, 10%.

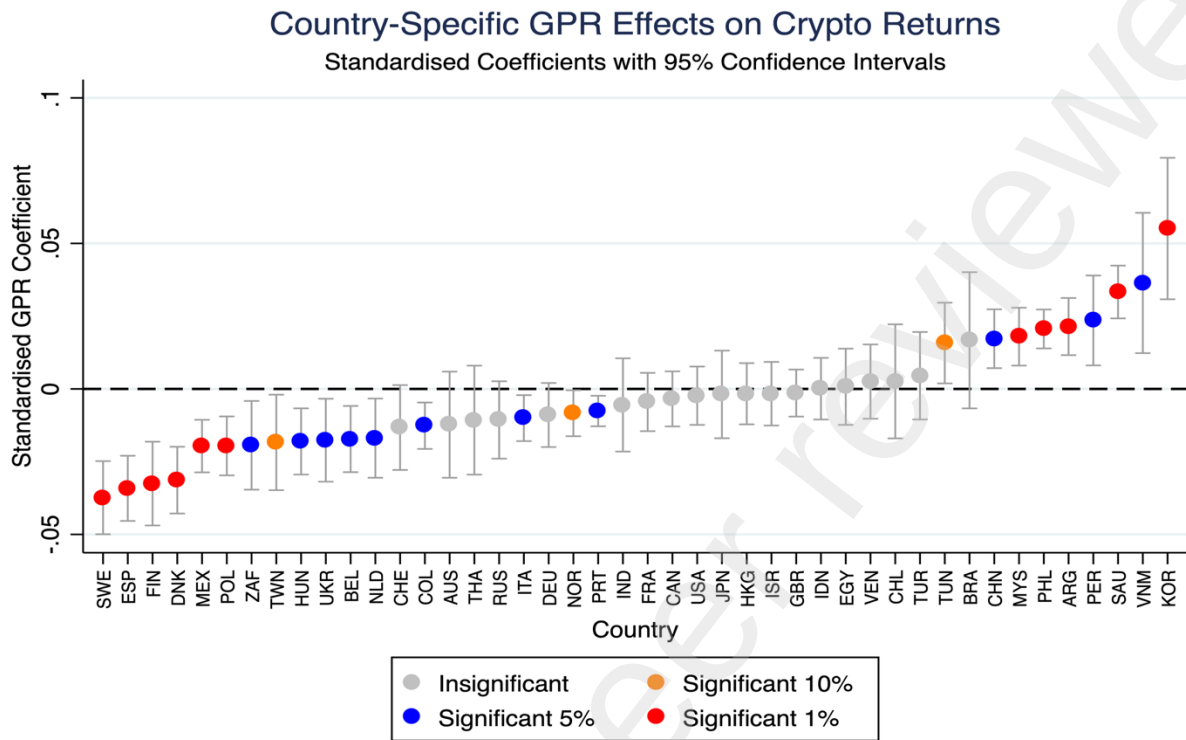
Table 5: Granger Causality-in-Quantiles Test

H₀: A change in marginal GPR does not Granger-cause a change in cryptocurrency returns (Umar et al., 2022)

Cryptocurrency	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
ADA	0.314 [0.905]	0.349 [0.883]	1.885 [0.094*]	2.073 [0.066*]	2.309 [0.042**]
BCH	1.187 [0.313]	0.423 [0.833]	0.399 [0.850]	2.273 [0.045**]	0.914 [0.471]
BNB	3.183 [0.007***]	1.314 [0.255]	0.548 [0.740]	3.984 [0.001***]	3.593 [0.003***]
BTC	0.575 [0.720]	0.390 [0.856]	1.044 [0.390]	3.863 [0.002***]	3.350 [0.005***]
DOGE	0.192 [0.966]	1.235 [0.290]	1.487 [0.191]	1.398 [0.222]	1.204 [0.305]
ETH	0.813 [0.540]	0.291 [0.918]	0.700 [0.623]	3.156 [0.008***]	2.923 [0.012**]
TRON	4.128 [0.001***]	5.781 [0.000***]	0.869 [0.501]	6.158 [0.000***]	6.646 [0.000***]
XRP	1.152 [0.331]	0.873 [0.498]	0.440 [0.821]	1.366 [0.234]	1.151 [0.331]

*Note: Table reports F-statistics with p-values in brackets. F-statistics are from joint significance tests of five GPR lags in quantile regressions estimated separately for each cryptocurrency, following Umar et al. (2022). The null hypothesis is that a change in marginal GPR does not Granger-cause a change in cryptocurrency returns. Three lags of returns are included as controls. $\tau = 0.10$ and 0.25 represent bearish conditions; $\tau = 0.50$ normal conditions; $\tau = 0.75$ and 0.90 bullish conditions. ***, **, * denote significance at 1%, 5%, and 10% levels respectively.*

Figure 1: Impact of Country-level GPR on Crypto Returns



Note: Coefficients sorted by magnitude. Red = 1%, Blue = 5%, Orange = 10%, Grey = Insignificant

Figure 2: Impact of GPR on Crypto Returns (GPR, GPRA, GPRT)

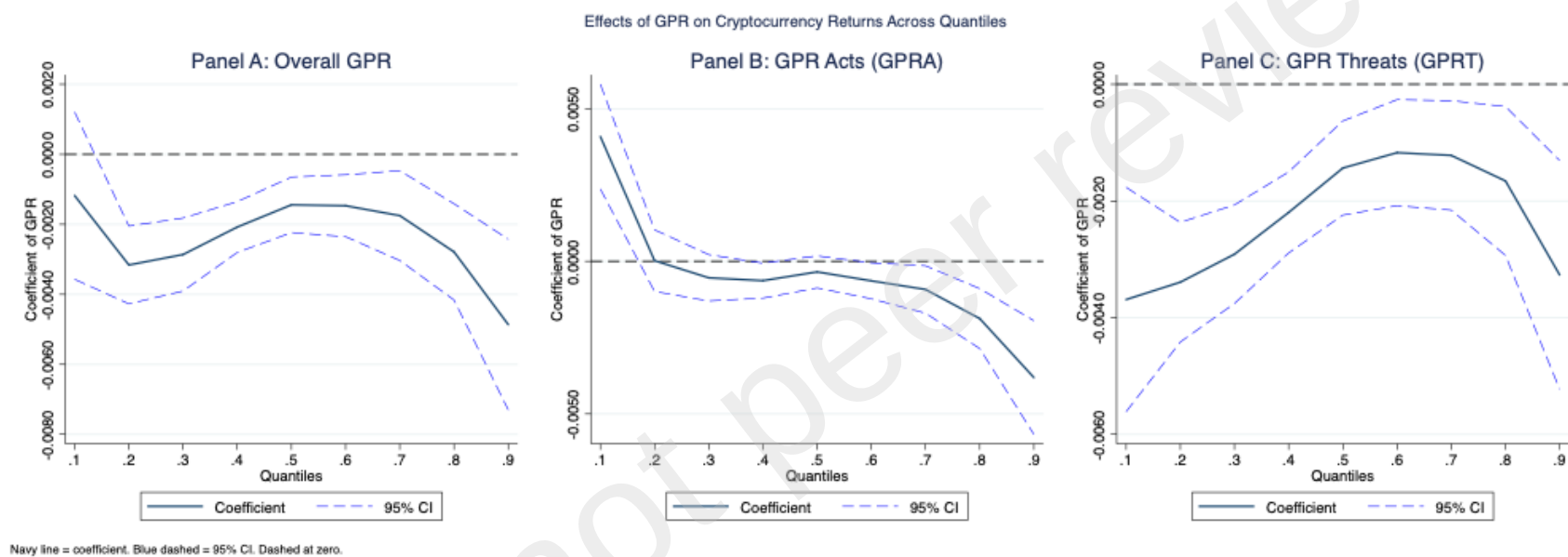
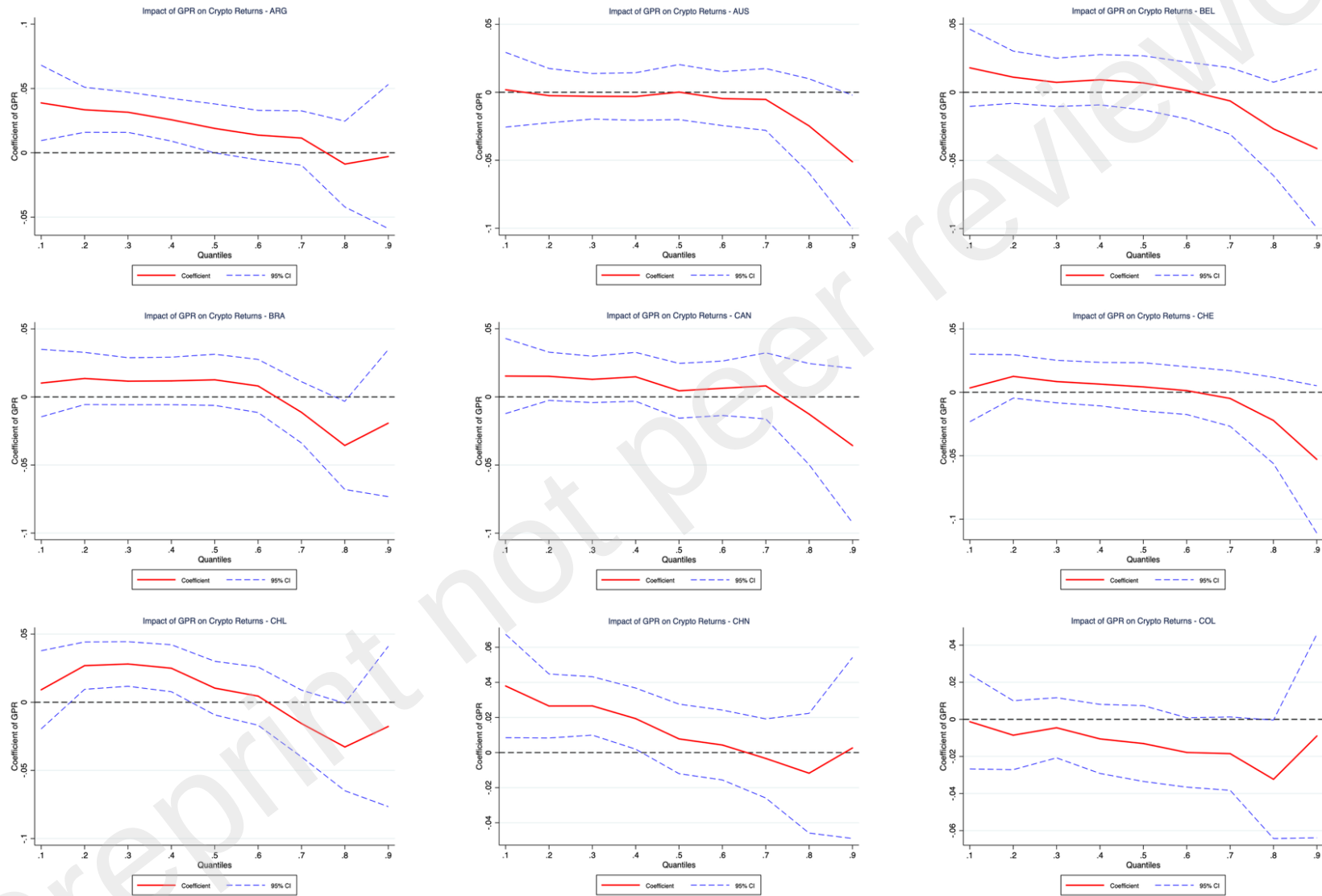
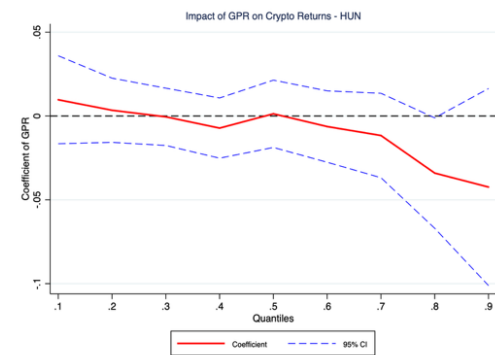
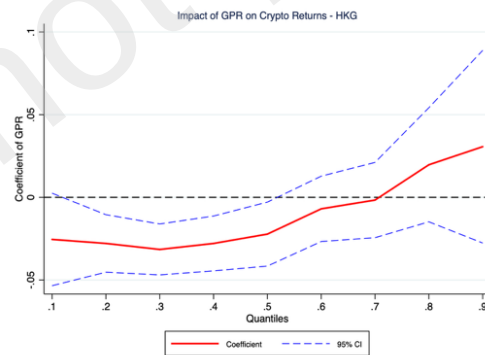
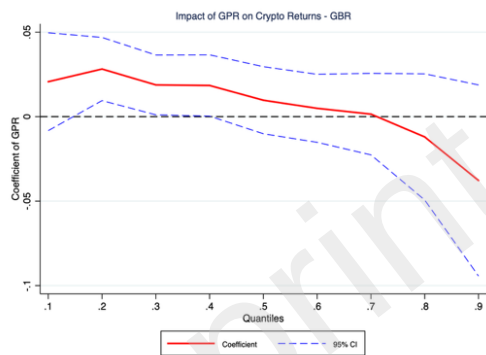
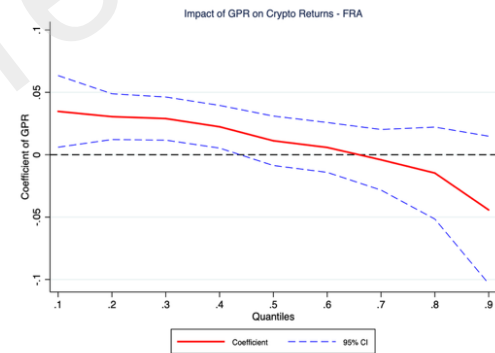
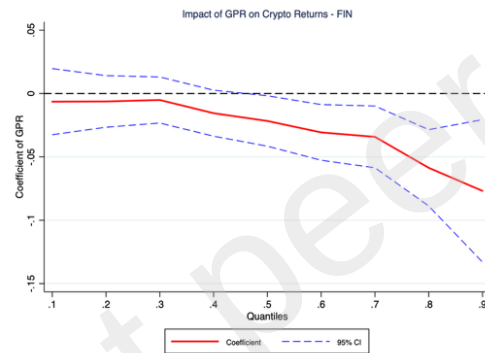
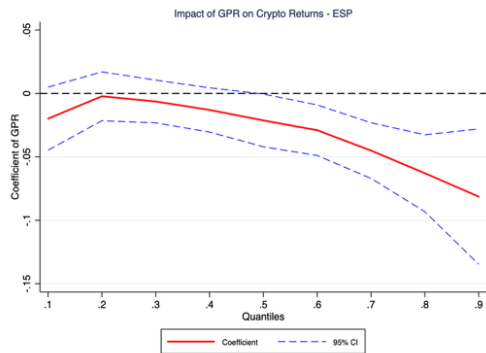
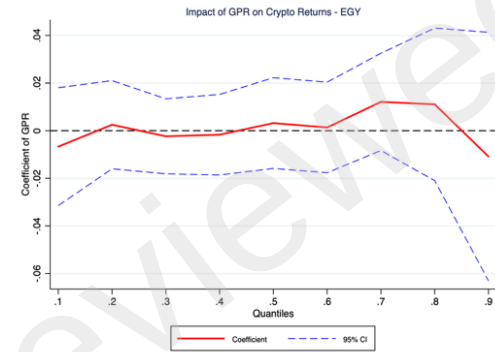
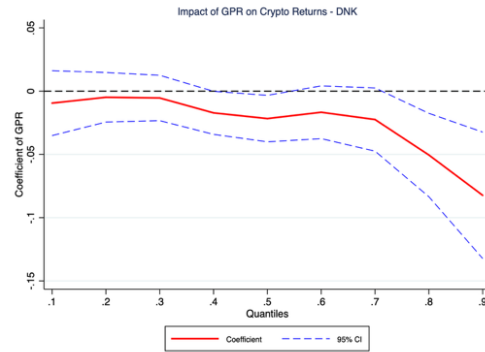
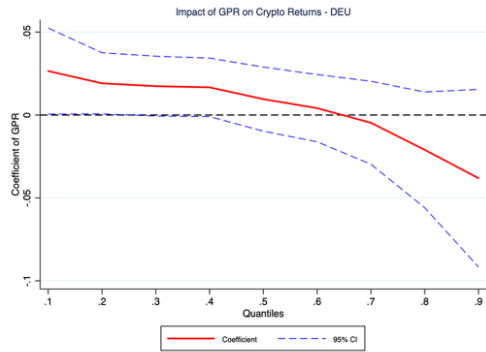
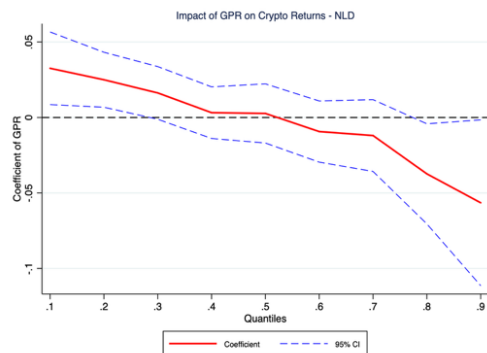
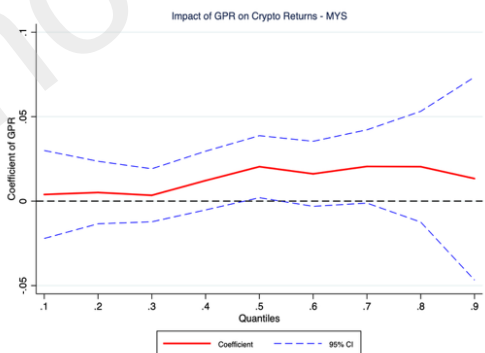
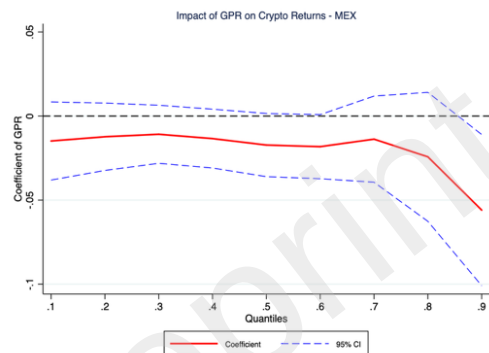
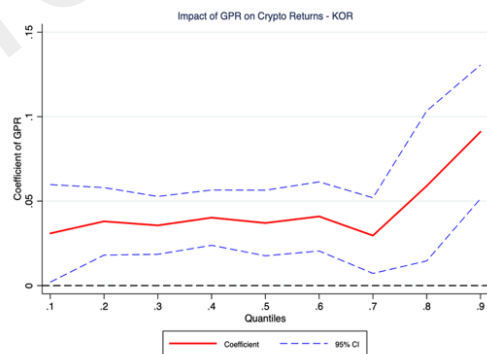
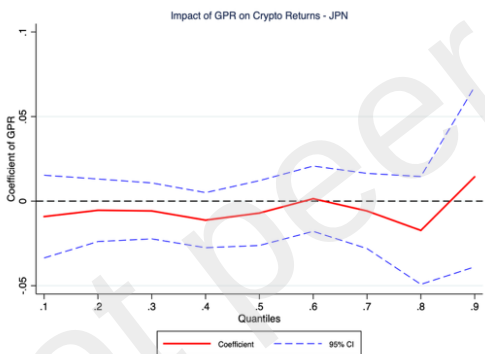
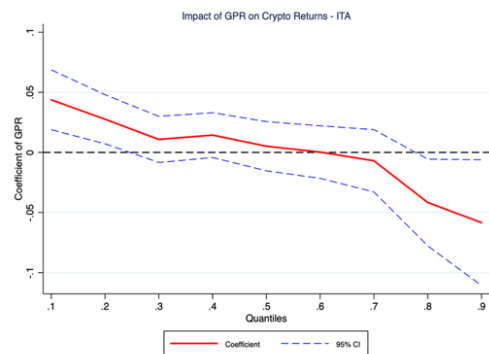
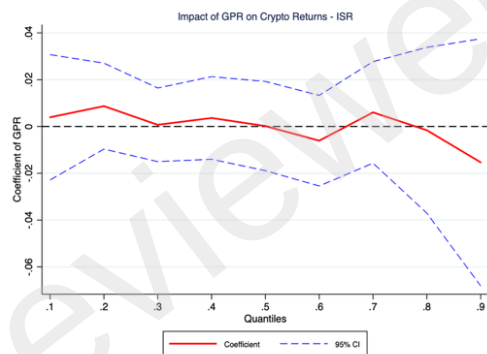
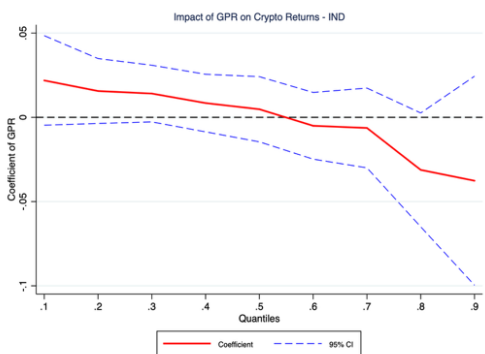
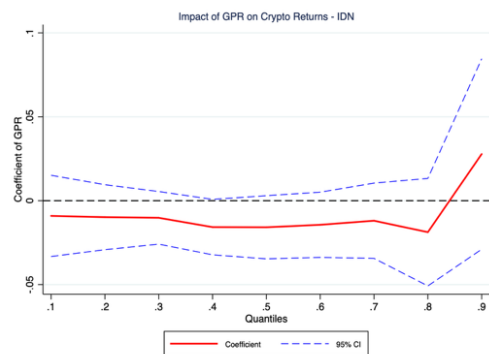
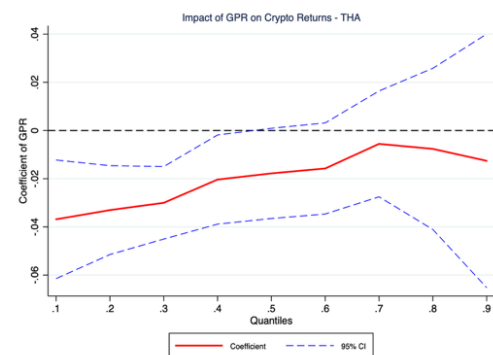
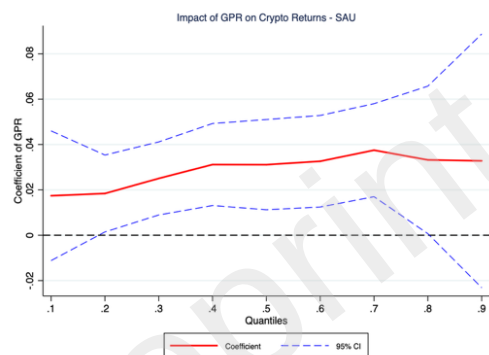
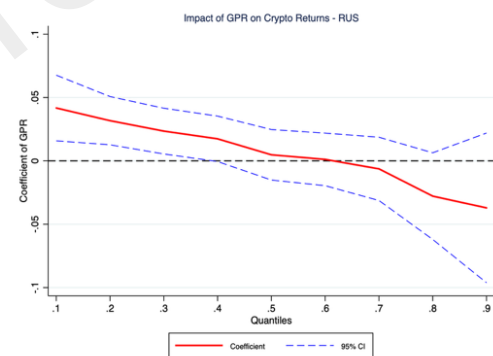
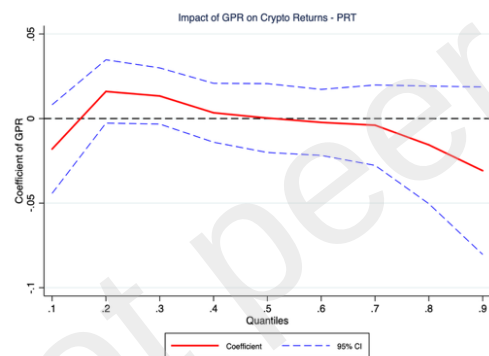
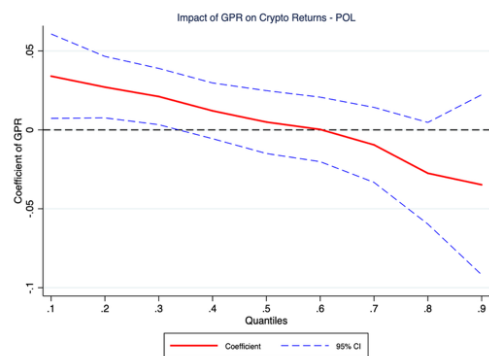
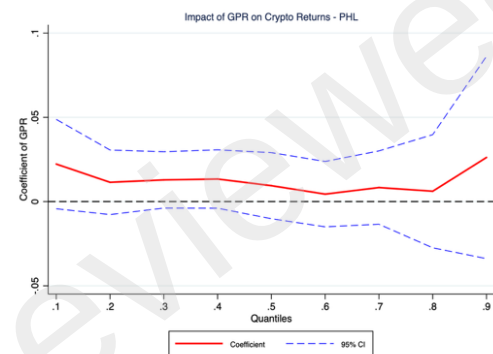
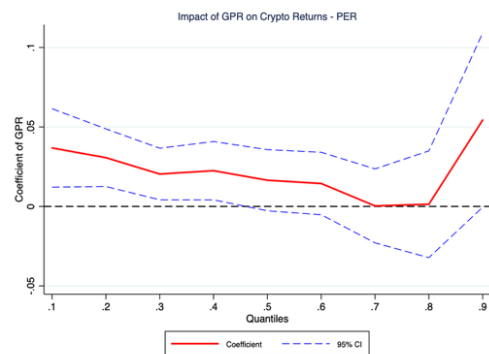


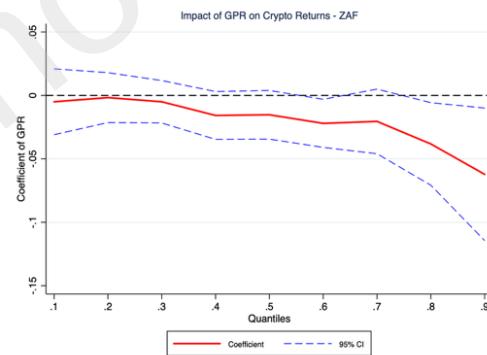
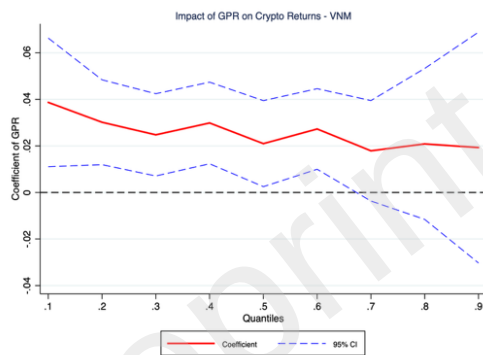
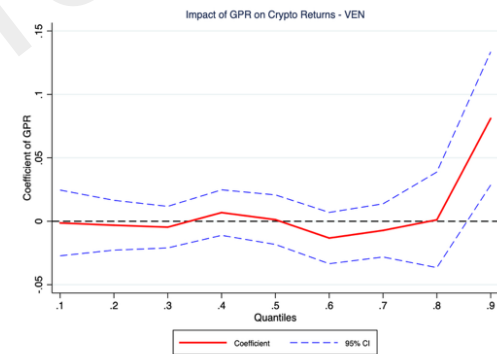
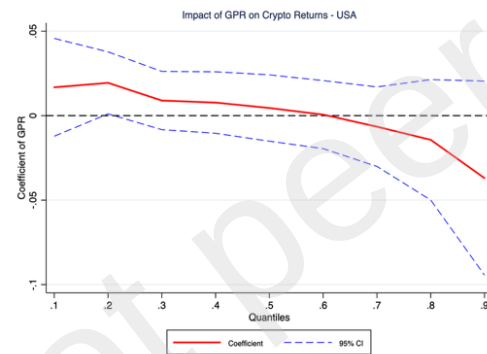
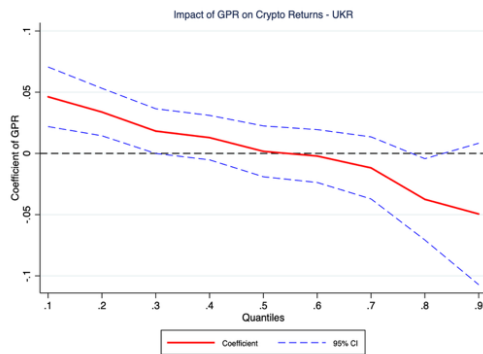
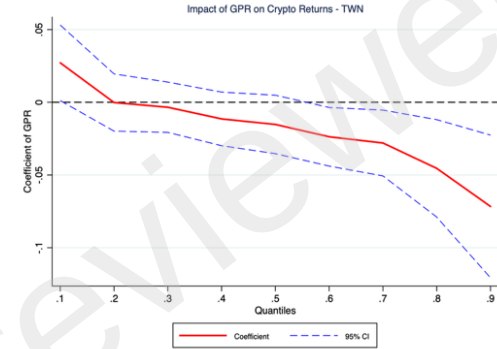
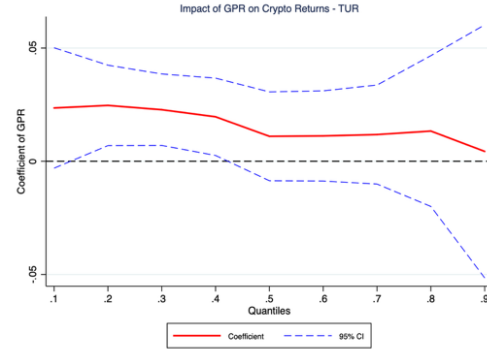
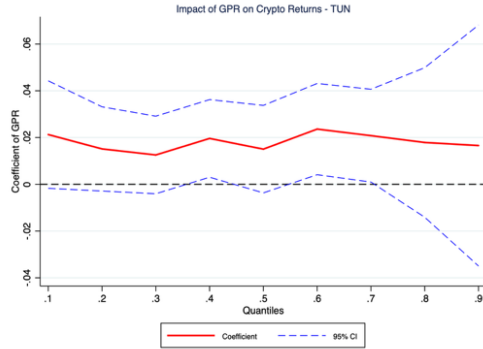
Figure 3: Country-wise QR plots on the impact of GPR on crypto returns (with recent GPR index)











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